

Graphical Quadratic Algebra: a Complete Calculus for Convex Optimisation and Gaussian Probability

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ABSTRACT

Convex analysis and Gaussian probability are tightly connected, as mostly evident in the theory of linear regression. Our work introduces an algebraic perspective on this relationship, in the form of a diagrammatic calculus of string diagrams, called Graphical Quadratic Algebra (GQA). We show that GQA is a complete axiomatisation for the category of quadratic relations, a compositional formulation of quadratic optimisation problems. Moreover, we identify a sub-theory of GQA which is complete for the category of Gaussian probabilistic processes. We show how GQA may be used to study linear regression, probabilistic programming, and electrical circuits.

1. Introduction

Gaussian probability theory studies phenomena governed by *normal* distributions — bell-shaped curves fully determined by mean and variance. Quadratic optimisation, on the other hand, is concerned with minimising quadratic functions — those expressible as polynomials of degree at most 2. Though they arise in different contexts, these two areas are deeply related. A clear instance of this connection appears in linear regression, where one seeks the best-fitting solution to a linear system $Ax = b$, with A a matrix and b a vector representing observed data. If the error term ϵ is assumed to follow a Gaussian distribution, then finding the most likely solution to the model $Ax + \epsilon = b$ is equivalent to minimising the quadratic loss function $f(x) = \|Ax - b\|^2$, which measures the sum of squared residuals. The relationship goes further: the log-density of a Gaussian distribution is itself a quadratic function, and computing conditional distributions in Gaussian models often reduces to solving least-squares problems.

In this paper, we introduce a fresh perspective on Gaussian probability, quadratic optimisation, and their web of connections, by presenting them as categorical structures. Our main results are *complete axiomatisations* for these categories, via an axiomatic calculus called *graphical quadratic algebra*.

Our approach unfolds as follows: first, we study how quadratic problems *compose*, so that they can be organised into a category. To this aim, we view a partial quadratic function $F : \mathbb{R}^m \times \mathbb{R}^n \rightarrow [0, \infty]$ as a $[0, \infty]$ -weighted relation on $\mathbb{R}^m \times \mathbb{R}^n$, called a *quadratic relation*. This perspective has three major appeals. First, quadratic relations characterise the solutions of constrained least-square problems (Boyd and Vandenberghe (2004)). Second, their composition is naturally defined via constrained minimisation (see (10) below). Third, quadratic relations extend affine relations (Bonchi, Piedeleu, Sobocinski and Zanasi (2019)), linking our work with the theory of *Graphical Affine Algebra* (Bonchi et al. (2019)) and offering a clear pathway towards a complete axiomatic calculus.

The resulting category **QuadRel**, with morphisms the quadratic relations, provides an algebraic setting to study quadratic optimisation. Gaussian probability enters the picture as a subcategory **Gauss** of **QuadRel**. The category **Gauss** has been previously studied as a *Markov category*, in the context of a more general framework for categorical probability theory Fritz (2020). Morphisms of **Gauss**, called Gaussian stochastic maps, generalise Gaussian distributions in the same way as linear maps generalise vectors. The study of **Gauss** is of independent interest because of its use as a semantics for Gaussian probabilistic programming (Stein and Staton (2021)).

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Once the ‘semantic’ structures **QuadRel** and **Gauss** are in place, we introduce their ‘syntactic’ presentation. Our language adopts the graphical notation of *string diagrams* (see eg. Selinger (2011); Piedeleu and Zanasi (2025) for recent introductions). As a two-dimensional formalism, string diagrams can be flexibly treated both as syntax and as combinatorial objects. They are now widely adopted in axiomatic reasoning on resource-sensitive models, eg. in linear algebra (Bonchi, Sobociński and Zanasi (2017); Coecke, Genovese, Lewis, Marsden and Touni (2018); Baez and Erbele (2015); Coecke and Kissinger (2018)) and probabilistic reasoning (Fritz (2020); Jacobs, Kissinger and Zanasi (2021); Lorenz and Tull (2023); Jacobs and Zanasi (2020)), making it a natural choice for our approach.

The language we introduce, called **GQA** (‘Graphical Quadratic Algebra’), consists of the string diagrams obtained by sequential and parallel composition of the following generators, for $k \in \mathbb{R}$:

(1)

As the name suggests, **GQA** directly extends *Graphical Affine Algebra* (**GAA**, Bonchi et al. (2019)), which is given by the generators of the first two blocks in (1). Semantically, the generators in the first block model basic linear algebraic operations of copying ($\text{---}\bullet\text{---}$), discarding ($\text{---}\bullet$), addition ($\text{---}\cup\text{---}$), scaling ($\text{---}[k]\text{---}$), and the constants zero ($\bigcirc$) and one ($|$). With the equational theory of Hopf bimonoids (Figure 1, first block), string diagrams built with these generators are known to axiomatise the category **AffVect** of affine maps (Bonchi et al. (2019)). If we add the generators of the second block of (1), namely $\text{---}\bullet\text{---}$, $\text{---}\bullet$, $\text{---}\cup\text{---}$, $\text{---}\cap\text{---}$, with behaviour symmetric to their mirrored version and extra equations (third block in Figure 1, notably including two Frobenius monoid structures), we obtain the full **GAA**, which axiomatises the category **AffRel** of affine relations, i.e. affine subspaces that compose relationally (Bonchi et al. (2019)).

The surprising discovery of our work is that, in order to extend the axiomatisation of the category **AffRel** of affine relations to one of the category **QuadRel** of *quadratic* relations, it suffices to add just one extra generator, $\leftarrow$ (and three equations as in Figure 1, second block): this yields **GQA**. Moreover, we axiomatise **Gauss** by restricting **GQA** to generators in the first and third block in (1).

In **Gauss**, $\leftarrow$ is interpreted as the standard normal distribution $\mathcal{N}(0, 1)$, and in **QuadRel** as the quadratic function $f(x) = \frac{1}{2}x^2$ (ie., a unary quadratic relation on $\mathbb{R}$). We illustrate this dual interpretation through the worked example of Section 2.

Completeness of **GQA** for **QuadRel** and **Gauss** ensures that any semantic equality holding in **QuadRel** or **Gauss** can in principle be derived. More broadly, this algebraic characterisation sheds light on the connections between quadratic optimisation and Gaussian probability theory, revealing their relative expressive power. Notably, the equational theory of **Gauss** is a fragment of the one of **QuadRel**, without the ability to model relational behaviours using Frobenius monoids. Moreover, the presentation of **Gauss** omits the generator $\text{---}\cap\text{---}$, which we may regard as conditioning of Gaussians. From this perspective, we may view quadratic relations as generalised Gaussian distributions, which combine probability and conditioning.

There are a few more interesting ramifications of our work. As noted, there are transformations relating Gaussian distributions, quadratic problems, and affine spaces. However, turning them into functors is technically challenging, as studied in Stein and Samuelson (2024). Thanks to our axiomatic presentations, it suffices to define these mappings on the generators of **GQA** to obtain functoriality as a simple corollary. We explore this in Section 6.1.

Our characterisations also enable a new methodology, based on equational diagrammatic reasoning, to study domains modelled by quadratic relations or Gaussian distributions, such as linear regression, noisy electrical circuits (linking our approach to Willems’ framework of open stochastic systems in control theory, e.g. Willems (2013)) and probabilistic program semantics. We explore these examples in Section 6. As a preview, the program below left is written in a language with support for Gaussian distributions and a conditioning operator ($\text{---}\text{=}\text{=}\text{---}$), and infers the distribution of a latent variable from a noisy observation. The corresponding **GQA**-diagram is below right.

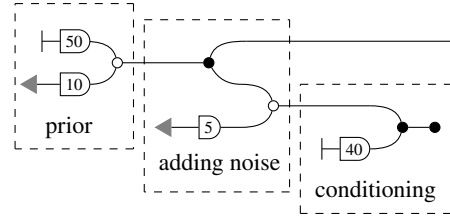
Calculus	Presents ($\cong$)	Morphisms	Reference
$\overline{\mathbf{GAA}}$	AffVect	affine maps	Bonchi et al. (2019)
$\overline{\mathbf{GQA}}$	Gauss	Gaussian maps	Theorem 4.3
$\overline{\mathbf{GQA}}_E$	GaussEx	extended Gaussian maps	Theorem A.14
$\mathbf{GAA}$	AffRel	affine relations	Bonchi et al. (2019)
$\mathbf{GQA}$	QuadRel	quadratic relations	Theorem 5.11

Table 1

The string diagrammatic calculi of this paper and the props they present, each via an isomorphism ($\cong$); the affine cases are due to Bonchi et al. (2019). The calculi form two chains of inclusions, $\overline{\mathbf{GAA}} \subseteq \overline{\mathbf{GQA}} \subseteq \overline{\mathbf{GQA}}_E \subseteq \mathbf{GQA}$ and $\overline{\mathbf{GAA}} \subseteq \mathbf{GAA} \subseteq \mathbf{GQA}$, matched by the semantic inclusions $\mathbf{AffVect} \subseteq \mathbf{Gauss} \subseteq \mathbf{GaussEx} \subseteq \mathbf{QuadRel}$ and $\mathbf{AffVect} \subseteq \mathbf{AffRel} \subseteq \mathbf{QuadRel}$. Each extension is obtained by adding generators from (1): passing from $\overline{\mathbf{GAA}}$ to $\overline{\mathbf{GQA}}$ adds a single new generator (the quadratic one), while the full calculi additionally carry the relational (Frobenius) structure.

```

let value = 50 + 10 · normal() in
let measurement = value + 5 · normal() in
measurement := 40;
value
    
```



Programs in this language correspond systematically to string diagrams. Moreover, the axioms of $\mathbf{GQA}$ suffice to reduce such a diagram to one representing the posterior distribution, effectively mirroring the program's execution.

Synopsis Table 1 gives an overview of the categories studied in this paper and the calculi presenting them. Section 2 works through a motivating example that previews the whole development. Section 3 recalls background material on string diagrammatic calculi for linear algebra, with additional preliminaries found in Appendix A.1. Then we present our work in two steps: first we introduce the simpler category **Gauss**, and show how the fragment $\overline{\mathbf{GQA}}$ axiomatises it (Section 4). Next, we introduce **QuadRel** and its axiomatic calculus, the full $\mathbf{GQA}$ (Section 5). Details of the various completeness proofs are found in Appendix A.2, A.4, and A.5. In Section 6 we cover some applications of our framework: functorial transformations (Section 6.1), linear regression (Section 6.2), probabilistic programming (Section 6.4), and noisy electrical circuits (Section 6.6). We then conclude by outlining future work (Section 7).

Conference Version This work is a substantial extension of the conference paper Stein, Zanasi, Piedeleu and Samuelson (2025) presented at the *International Colloquium on Theoretical Aspects of Computing* (ICTAC). It provides all the missing proofs, including details of the various completeness arguments, additional background on diagrammatic calculi (Section 3 and Section A.1), and a more articulated motivating example (Section 2). The case studies of Section 6.2 and Section 6.4 have been significantly expanded. Finally, the metatheory of conditioning (Section 6.3), the case study of Section 6.6, the connections to resistor networks and symplectic algebra (Section 6.5), and the completeness result for extended Gaussian processes (Section A.4) are new compared to Stein et al. (2025).

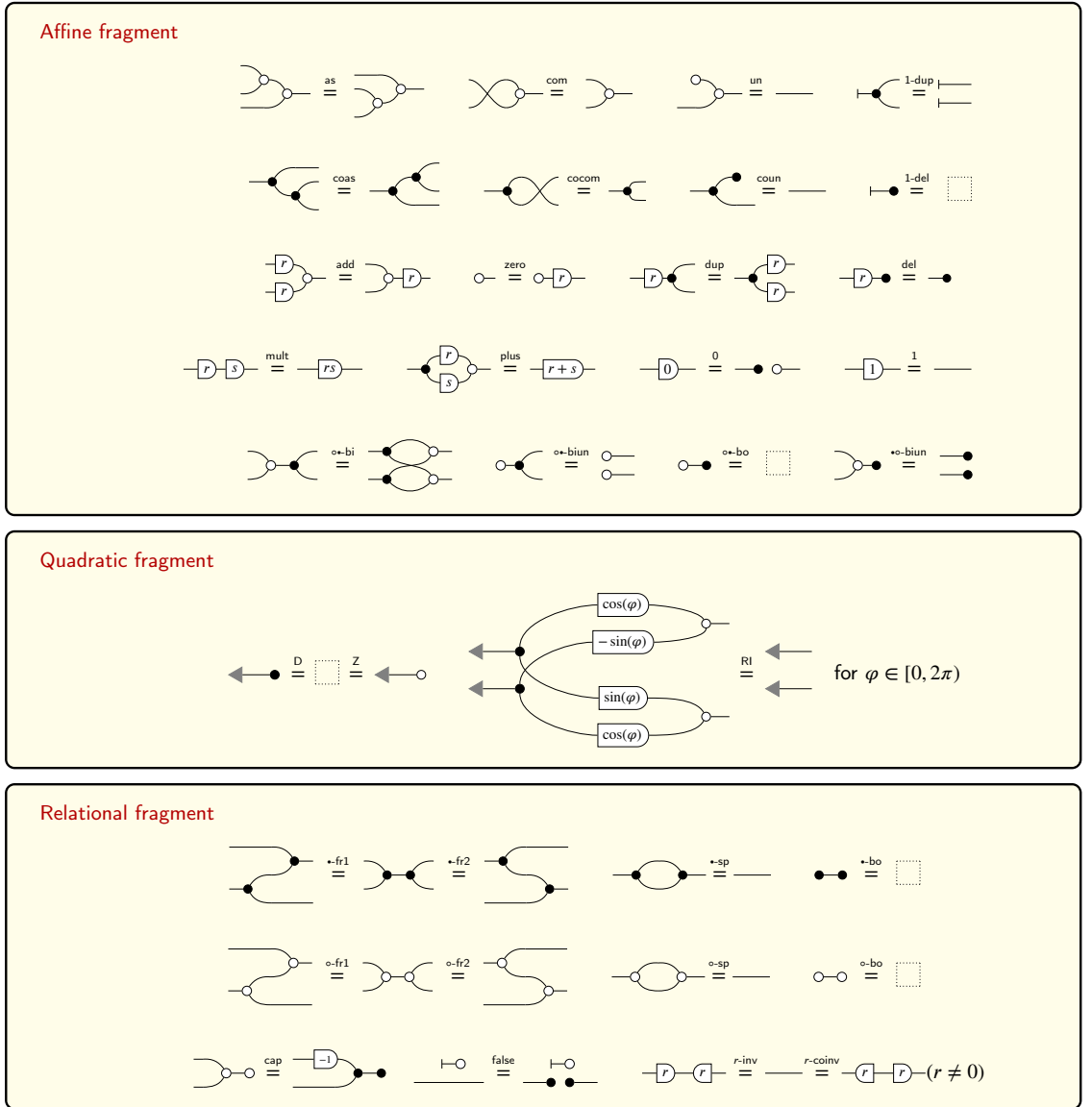


Figure 1: Axioms of Graphical Quadratic Algebra

2. A Motivating Example

We preview the contributions of the paper through a single, elementary example: two seemingly unrelated problems—one about Gaussian distributions, one a constrained least-squares optimisation—that are solved by the same rotation-invariance argument, and that the calculus **GQA** renders as the same equational derivation between string diagrams. The two problems are as follows:

Problem A If $X_1, X_2 \sim \mathcal{N}(0, 1)$ are independent, show that their sum $X_1 + X_2 \sim \mathcal{N}(0, 2)$.

Problem B Show that the constrained minimisation problem

$$\inf \left\{ \frac{1}{2}x_1^2 + \frac{1}{2}x_2^2 : x_1 + x_2 = y \right\} \text{ has solution } \frac{1}{4}y^2. \quad (2)$$

The first problem can be solved by computing the integral convolution of Gaussian probability density functions

$$\int \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}x^2\right) \cdot \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{1}{2}(y-x)^2\right) dx = \frac{1}{\sqrt{4\pi}} \exp\left(-\frac{1}{4}y^2\right)$$

which results in the density of $\mathcal{N}(0, 2)$. The second problem can be solved by noting that the function $x \mapsto \frac{1}{2}x^2 + \frac{1}{2}(y-x)^2$ has vanishing derivative at $x = \frac{1}{2}y$. Our work is about a unified, abstract semantics for both of these problems. We give an outline for how to arrive there in three steps:

The **first step** that hints towards a connection between these problems is that the quadratic functions which appear in Problem B are, up to an additive constant, the negative log-densities of the Gaussians in Problem A—that is, the functions $-\log f$ obtained from a Gaussian density f , a correspondence we make precise in Section 6.1. Further, the expression (2) is also a kind of convolution, called an infimal convolution, of quadratic functions.

It appears that the function $-\log$ which maps a Gaussian density to its negative log also turns integral convolutions into inf-convolutions. This is not the case for arbitrary functions, but it *is* the case for Gaussian densities, specifically. Gaussians can therefore be treated in either the probabilistic world or the world of convex optimisation, to the same effect.

If we view the two different kinds of convolution as a particular way to *compose* Gaussians or quadratic functions, respectively, our observation means that $-\log$ preserves this composition, i.e. becomes *functorial*. The precise way to set up Gaussians and quadratic functions as categories **Gauss** and **QuadRel**, and what their composition means, is nontrivial and part of the contribution of this work. Roughly:

1. **Gauss** will consist of stochastic linear maps with Gaussian noise, and composition is function composition and independent accumulation of the noise;
2. **QuadRel** will consist of partial quadratic functions, and composition is constrained infimisation.

By Stein and Samuelson (2024), $-\log$ indeed defines a functorial embedding **Gauss** $\rightarrow$ **QuadRel**, but the proof is complex. This work goes further in finding an elegant and abstract reason for this functorial relationship to exist.

The **second step** is to realise that there is a common proof strategy which allows us to tackle both problems:

1. The joint distribution (X_1, X_2) is invariant under rotations in the plane. This is a specific feature of Gaussian distributions; indeed the *Herschel-Maxwell theorem* says that *only* Gaussian distributions have this property (e.g. Gyenis (2017)). So to compute the distribution of $X_1 + X_2$, we may compute the distribution of $U_1 + U_2$ where

$$(U_1, U_2) = \left(\frac{1}{\sqrt{2}}X_1 - \frac{1}{\sqrt{2}}X_2, \frac{1}{\sqrt{2}}X_1 + \frac{1}{\sqrt{2}}X_2 \right)$$

is the vector (X_1, X_2) rotated by 45 degrees. But then we simplify

$$U_1 + U_2 = \frac{1}{\sqrt{2}}X_1 + \frac{1}{\sqrt{2}}X_1 = \sqrt{2} \cdot X_1$$

from which we obtain $U_1 + U_2 \sim \mathcal{N}(0, 2)$ by simply rescaling the distribution of X_1 appropriately.

2. The infimand $\frac{1}{2}x_1^2 + \frac{1}{2}x_2^2$ in the constrained optimisation problem is also invariant under rotations; applying the same coordinate change as before, we obtain the transformed problem

$$\inf \left\{ \frac{1}{2}u_1^2 + \frac{1}{2}u_2^2 : u_1 + u_2 = y \right\} = \inf \left\{ \frac{1}{2}x_1^2 + \frac{1}{2}x_2^2 : u_1 + u_2 = y \right\} = \inf \left\{ \frac{1}{2}x_1^2 + \frac{1}{2}x_2^2 : \sqrt{2} \cdot x_1 = y \right\}$$

which is clearly solved by $x_1 = \frac{1}{\sqrt{2}}y$ and $x_2 = 0$, giving the desired answer

$$\frac{1}{2} \left(\frac{1}{\sqrt{2}}y \right)^2 + \frac{1}{2}0^2 = \frac{1}{4}y^2$$

We note that this result, too, is an appropriate rescaling of the original quadratic function $x_1 \mapsto \frac{1}{2}x_1^2$.

The **third step** is to generalise this technique and make it fully formal. It turns out that rotation invariance *fully characterises* both Gaussian distributions and quadratic optimisation problems in a certain algebraic sense. To make this precise, we devise an algebraic theory, called **GQA** for *Graphical Quadratic Algebra*, which has two intended models: Gaussian distributions and quadratic optimisation problems. In this theory, rotation invariance becomes simply an axiom. Terms in this theory can be rendered as *string diagrams*, so they can be displayed and manipulated in an intuitive and appealing way. Our two problems A and B are now represented by the very same equation, which we show is derivable in **GQA**:

$$\begin{array}{c} \text{---} \circ \text{---} \\ \text{---} \circ \text{---} \end{array} = \begin{array}{c} \text{---} \circ \text{---} \\ \text{---} \circ \text{---} \end{array} \boxed{\sqrt{2}}$$

We will explain later what this means in detail. Both sides of this equation are *states*, i.e. diagrams with no inputs and a single output (morphisms $0 \rightarrow 1$), so each denotes a quadratic relation on $\mathbb{R}^0 \times \mathbb{R}^1$. With this in mind, these diagrams have a meaning that can be instantiated in either Gaussians or quadratic optimisation problems:

1. In **Gauss**, the grey triangle $\triangleleft$ represents the standard normal distribution, and $\text{---} \circ \text{---}$ is addition. Composition is pushforward. Hence the left-hand side represents the sum $\mathcal{N}(0, 1) + \mathcal{N}(0, 1)$, and the right-hand side the rescaling $\sqrt{2} \cdot \mathcal{N}(0, 1) = \mathcal{N}(0, 2)$.
2. In **QuadRel**, $\triangleleft$ represents the quadratic function $x \mapsto \frac{1}{2}x^2$, and $\text{---} \circ \text{---}$ is the constraint $\{x_1 + x_2 = y\}$. Composition is infimisation over the joint variable, so the left-hand side becomes

$$\inf \left\{ \frac{1}{2}x_1^2 + \frac{1}{2}x_2^2 : x_1 + x_2 = y \right\} \text{ while the right-hand side becomes } \inf \left\{ \frac{1}{2}x^2 : \sqrt{2} \cdot x = y \right\} = \frac{1}{4}y^2$$

We elaborate how this semantics is formally computed in Examples 5.1 and 5.8.

We can now replay the proof steps for solving problems A and B internally to the theory **GQA**:

$$\begin{array}{c} \text{---} \circ \text{---} \\ \text{---} \circ \text{---} \end{array} \stackrel{\text{RI}}{=} \begin{array}{c} \text{---} \circ \text{---} \\ \text{---} \circ \text{---} \end{array} \begin{array}{c} \boxed{1/\sqrt{2}} \\ \boxed{-1/\sqrt{2}} \\ \boxed{1/\sqrt{2}} \\ \boxed{1/\sqrt{2}} \end{array} \stackrel{\text{GAA}}{=} \begin{array}{c} \text{---} \circ \text{---} \\ \text{---} \circ \text{---} \end{array} \boxed{-1} \begin{array}{c} \text{---} \circ \text{---} \\ \text{---} \circ \text{---} \end{array} \boxed{1/\sqrt{2}} \stackrel{\text{GAA}}{=} \begin{array}{c} \text{---} \circ \text{---} \\ \text{---} \circ \text{---} \end{array} \boxed{\sqrt{2}} \stackrel{\text{D}}{=} \begin{array}{c} \text{---} \circ \text{---} \\ \text{---} \circ \text{---} \end{array} \boxed{\sqrt{2}} \quad (3)$$

The crucial first step is applying the rotation invariance axiom RI using the same coordinate change as before, now suitably encoded as a string diagram internal to the theory. The steps labelled GAA use the equations of *Graphical Affine Algebra* to perform some linear-algebraic reasoning inside the theory, which amounts to checking the matrix equation

$$(1 \quad 1) \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} = (\sqrt{2} \quad 0) \quad (4)$$

A detailed equational derivation of these steps is given in Example 3.3.

The final step D is a discardability axiom. How to axiomatise linear algebra is well-known (e.g. Paixao, Rufino and Sobocinski (2022); Bonchi et al. (2019)) and not the focus of this article, though we will briefly recall it in Section 3. Our goal is to show how, by the mere addition of $\leftarrow$ and axioms $\{R, D, Z\}$ to these theories, we obtain **complete theories** of both Gaussians and quadratic optimisation.

3. Background: Graphical Calculi for Linear Algebra

We now recall background material on string diagrams and Graphical Affine Algebra, referring to Bonchi et al. (2019) for more details. First, we introduce the notion of a *prop*, which is the categorical setting for our work. A prop (MacLane (1965)) is a symmetric strict monoidal category with set of objects the natural numbers, and monoidal product on objects given by addition. Prop morphisms are identity-on-objects symmetric monoidal functors between props. Props play a role akin to algebraic clones, in a monoidal setting. The linear algebraic transformations that are the subject of this paper may be organised into props.

Definition 3.1. *The prop **AffVect** has affine maps of type $\mathbb{R}^m \rightarrow \mathbb{R}^n$ as morphisms $m \rightarrow n$, with composition given by matrix multiplication and monoidal product given by direct sum. The prop **AffRel** has affine subspaces of $\mathbb{R}^{m+n}$ as morphisms $m \rightarrow n$, with relational composition R ; $S = \{(\mathbf{v}, \mathbf{w}) \mid \exists \mathbf{u}. (\mathbf{v}, \mathbf{u}) \in R \wedge (\mathbf{u}, \mathbf{w}) \in S\}$, and monoidal product given by direct sum.*

In both props the object n stands for the space $\mathbb{R}^n$; in particular, the monoidal unit is the object 0, i.e. the trivial vector space $\mathbb{R}^0 = \{0\}$ (and not $\mathbb{R}$, which is the object 1). Because of the way they compose, **AffRel**-morphisms are called *affine relations*.

Complete axiomatisations for these categories are provided by string diagrammatic calculi, which we now introduce. Before the formalities, it is worth recalling why such a characterisation is valuable. String diagrams are a well-established graphical notation for the morphisms of a prop (Joyal and Street (1991); Selinger (2011)), with wires representing the objects and boxes representing the morphisms of the category. Being two-dimensional, string diagrams put the two fundamental ways of composing morphisms in a prop, in *sequence* and in *parallel*, on an equal footing and display them at a glance. Equational reasoning then becomes an intuitive manipulation of diagrams, with the usual bookkeeping of variable names and bracketing absorbed into the geometry. This is especially valuable in *resource-sensitive* settings such as linear algebra, where values cannot in general be freely copied or discarded—precisely the situations that ordinary (cartesian) term syntax handles awkwardly. We refer to Piedeleu and Zanasi (2025) for a broader discussion.

Within our perspective, string diagrams play the role of a *syntax* for semantically-defined props, such as those of Definition 3.1.¹ In order to faithfully account for the semantics, string diagrammatic syntax needs to be further quotiented by equations: for instance, when string diagrams represent affine maps, equations need to identify diagrams representing equivalent maps. This is achieved via the notion of a *symmetric monoidal theory*, which we now introduce. A symmetric monoidal theory (SMT) is a pair (Σ, E) , where Σ is a signature of operations $o : m \rightarrow n$ with an arity m and coarity n , and E is a set of equations between Σ -terms. A Σ -term c of type $m \rightarrow n$ will be represented graphically as a *string diagram* (Piedeleu and Zanasi (2025); Selinger (2011)) with m dangling wires on the left and n on the right, also written $\overset{m}{\square} \overset{n}{\square}$. Formally, Σ -terms are freely obtained by sequential and parallel compositions of the operations in Σ together with the identity $\text{---} : 1 \rightarrow 1$, the symmetry $\bowtie : 2 \rightarrow 2$, and the ‘empty’ diagram $\square : 0 \rightarrow 0$. Sequential composition of Σ -terms $\overset{m}{\square} \overset{n}{\square}$ and $\overset{n}{\square} \overset{v}{\square}$ is depicted as $\overset{m}{\square} \overset{n}{\square} \overset{v}{\square}$, of type $m \rightarrow v$. Parallel composition of Σ -terms $\overset{m_1}{\square} \overset{n_1}{\square}$ and $\overset{m_2}{\square} \overset{n_2}{\square}$ is depicted as $\overset{m_1}{\square} \overset{m_2}{\square} \overset{n_1}{\square} \overset{n_2}{\square}$, of type $m_1 + m_2 \rightarrow n_1 + n_2$. We refer to Σ -terms quotiented by the laws of symmetric strict monoidal categories (Figure 2) as *string diagrams*.

Given an SMT (Σ, E) , the prop $\text{FreeP}_{(\Sigma, E)}$ freely generated by (Σ, E) has morphisms $m \rightarrow n$ the string diagrams of type $m \rightarrow n$ quotiented by E , with sequential and parallel composition defined as on the corresponding Σ -terms. We will typically regard $\text{FreeP}_{(\Sigma, E)}$ as a *string diagrammatic calculus*, with the string diagrams being its (graphical) *syntax*.

¹This ‘syntactic’ perspective determines the definition of string diagrams given below, which follows eg. Piedeleu and Zanasi (2025); Bonchi et al. (2019, 2017) and is more convenient for our axiomatic perspective. The reader may be familiar with the combinatorial (Bonchi, Gadducci, Kissinger, Sobocinski and Zanasi (2022)) and topological (Joyal and Street (1991)) view on string diagrams: the three are ultimately equivalent. Additional details may be found eg. in Piedeleu and Zanasi (2025).

The props defined ‘directly’, as **AffVect** and **AffRel**, are regarded as *semantic models* for diagrammatic calculi. When there is an isomorphism of props $\llbracket \cdot \rrbracket$ between $\text{FreeP}_{(\Sigma, E)}$ and another prop **C**, we say that **C** is *presented* (or axiomatised) by (Σ, E) . Because prop morphisms are identity-on-objects, to prove such result it suffices to prove that $\llbracket \cdot \rrbracket$ is full and faithful. In logical terms, we can phrase these requirements as:

- *soundness* (if $s = t$ in $\text{FreeP}_{(\Sigma, E)}$, then $\llbracket s \rrbracket = \llbracket t \rrbracket$);
- *completeness* (if $\llbracket s \rrbracket = \llbracket t \rrbracket$ in **C**, then $s = t$ in $\text{FreeP}_{(\Sigma, E)}$);
- *definability* (for every $f : m \rightarrow n$ in **C**, there exists a Σ -term $s : m \rightarrow n$ with $\llbracket s \rrbracket = f$).

We are now ready to introduce our first diagrammatic calculus. *Graphical Affine Algebra* is based on the following operations:

$$\begin{array}{cccccc}
 \begin{array}{c} \text{---} \bullet \text{---} \\ \text{---} \bullet \text{---} \end{array} & \begin{array}{c} \text{---} \bullet \\ \bullet \text{---} \end{array} & \begin{array}{c} \text{---} \circ \text{---} \\ \text{---} \circ \text{---} \end{array} & \begin{array}{c} \text{---} \circ \\ \text{---} \circ \end{array} & \begin{array}{c} \text{---} \boxed{k} \text{---} \\ \text{---} \boxed{k} \text{---} \end{array} & \begin{array}{c} \text{---} \text{---} \\ \text{---} \text{---} \end{array} & (5) \\
 \begin{array}{c} \text{---} \bullet \text{---} \\ \text{---} \bullet \text{---} \end{array} & \begin{array}{c} \text{---} \bullet \\ \bullet \text{---} \end{array} & \begin{array}{c} \text{---} \circ \text{---} \\ \text{---} \circ \text{---} \end{array} & \begin{array}{c} \text{---} \circ \\ \text{---} \circ \end{array} & \begin{array}{c} \text{---} \boxed{k} \text{---} \\ \text{---} \boxed{k} \text{---} \end{array} & \begin{array}{c} \text{---} \text{---} \\ \text{---} \text{---} \end{array} & (6)
 \end{array}$$

Row (5) denotes basic linear transformations such as copying ($\text{---} \bullet \text{---}$), discarding ($\text{---} \bullet$), addition ($\text{---} \circ \text{---}$), the constant zero ($\text{---} \circ$), multiplication by a scalar k ($\text{---} \boxed{k} \text{---}$), and the constant one ($\text{---} \text{---}$). A diagram built from these is read from left to right, as a data-flow diagram, by tracking the values carried on the wires; using the constant $\text{---} \text{---}$ we can encode any affine map, for example $f(x) = 2x + 5$ as $\text{---} \boxed{2} \text{---} \circ \text{---} \boxed{5} \text{---}$. Row (6) expresses the converse of these maps, regarded as relations (subspaces), with the ‘mirror’ of $\text{---} \boxed{k} \text{---}$ and $\text{---} \text{---}$ derivable as follows:

$$\begin{array}{c} \text{---} \boxed{k} \text{---} \end{array} := \begin{array}{c} \text{---} \bullet \text{---} \\ \bullet \text{---} \end{array} \quad \begin{array}{c} \text{---} \text{---} \end{array} := \begin{array}{c} \text{---} \bullet \text{---} \\ \bullet \text{---} \end{array}$$

The affine fragment in Figure 1 describes the interaction between operations in (5): $\text{---} \circ \text{---}$, $\text{---} \circ$ form a commutative monoid, $\text{---} \bullet \text{---}$, $\text{---} \bullet$ form a commutative comonoid, distributing over each other according to the laws of bimonoids. Furthermore, scalars and $\text{---} \text{---}$ distribute over the bimonoid, and there are laws expressing the field structure on $\mathbb{R}$.

The relational fragment in Figure 1 describes additional laws involving operations in (6) and $\text{---} \boxed{k} \text{---}$, $\text{---} \text{---}$ defined above: black and white special Frobenius monoids, the equivalence between the black and the white compact closed structure arising from the Frobenius algebras (cap), the fact that scalars have inverses ($r\text{-inv}$, $r\text{-coinv}$), and an ‘ex-falso quodlibet’ principle (false, where $\text{---} \circ$ expressed the ‘impossible’ condition that $1 = 0$). Note at this stage we may also prove that $\text{---} \circ \text{---}$, $\text{---} \circ$ form a commutative comonoid and $\text{---} \bullet \text{---}$, $\text{---} \bullet$ form a commutative monoid. We write $\overline{\mathbf{GAA}}$ for the prop freely generated by operations (5) and equations in the affine fragment. Similarly, we write $\mathbf{GAA}$ for the prop freely generated by operations (5)-(6), and equations in the causal and relational fragments. These props present affine maps and affine relations respectively.

Proposition 3.2 (Bonchi et al. (2019)). $\overline{\mathbf{GAA}}$ presents **AffVect** and $\mathbf{GAA}$ presents **AffRel**.

Importantly, Proposition 3.2 implies that, whenever we reason in the aforementioned calculi, any equivalence of matrices or subspaces represented by string diagrams is provable in the corresponding equational theory. This will allow us to use freely known facts of linear algebra (e.g. those recalled in Appendix A.1) in diagrammatic reasoning whenever needed, allowing us to focus on the quadratic extension that is the topic of this paper.

The fragment $\overline{\mathbf{GAA}}$ generated by Row (5) alone retains the input-output reading above, and is for this reason called the *causal fragment* (a notion we return to in Section 4); every such diagram is moreover *discardable*, in the sense that $\text{---} \boxed{A} \text{---} \bullet \text{---} \stackrel{\mathbf{GAA}}{=} \text{---} \bullet \text{---}$. In the full calculus $\mathbf{GAA}$ this directionality is lost: the generator $\text{---} \circ$ returns nothing but constrains its input to be zero (a form of conditioning), while $\text{---} \bullet \text{---}$ denotes the nondeterministic choice $\{x : x \in \mathbb{R}\}$. A diagram of $\mathbf{GAA}$ is provably equal to one of $\overline{\mathbf{GAA}}$ if and only if it is discardable and *copyable*, i.e. it commutes with copying $\text{---} \bullet \text{---}$.

In preparation for our completeness theorems, we recall standard notation to represent matrices and subspaces as string diagrams of $\mathbf{GAA}$. First, for readability, we use thick wires for an arbitrary number of ingoing/outgoing wires in string diagrams, as in $\text{---} \boxed{c} \text{---} := \text{---} \boxed{c} \text{---}$. We use a similar convention to represent multiple instances of the same

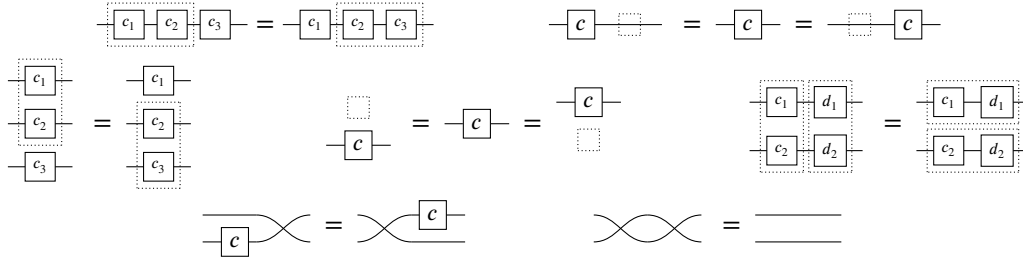


Figure 2: Laws of Symmetric Strict Monoidal Categories

4. Axiomatising Gaussian Probability

4.1. Gaussian Probability

Gaussian probability is an expressive self-contained fragment of probability theory. We recall what is strictly necessary here, referring to e.g. Billingsley (1995) for further background. A random variable X is called *normally distributed* or *Gaussian* with mean $\mu \in \mathbb{R}^n$ and variance σ^2 for $\sigma \in [0, \infty)$, if it has density function $f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$ with respect to the Lebesgue measure. This is typically written $X \sim \mathcal{N}(\mu, \sigma^2)$, meaning the distribution $\mathcal{N}(\mu, \sigma^2)$, also called Gaussian distribution, is the law of X . The *standard normal distribution* $\mathcal{N}(0, 1)$ is particularly significant, as it generates any other normal distribution, in the following sense: a random vector $\mathbf{X}$ is *multivariate normal* if its distribution arises as $\mathbf{X} = A \cdot (Z_1, \dots, Z_k) + \mu$, where $Z_1, \dots, Z_k \sim \mathcal{N}(0, 1)$ are independent variables and $A \in \mathbb{R}^{n \times k}$ is a matrix and $\mu \in \mathbb{R}^n$. In other words, multivariable normal distributions arise as the pushforwards of standard normal distributions under affine maps. The distribution of $\mathbf{X}$ is fully characterised by its mean μ and its covariance matrix $\Sigma = AA^T \in \mathbb{R}^{n \times n}$, for $A \in \mathbb{R}^{n \times k}$. Conversely, every positive semidefinite matrix Σ is the covariance matrix of a unique Gaussian distribution $\mathcal{N}(\mu, \Sigma)$, supported on the affine subspace $\mu + \text{im}(\Sigma)$. A convenient way of building up complicated Gaussian distributions is by pushing forward existing distributions under affine maps: if $\mathcal{N}(\mu, \Sigma)$ is a Gaussian distribution and A is a matrix, then the pushforward distribution $A_*\mathcal{N}(\mu, \Sigma)$ is $\mathcal{N}(A\mu, A\Sigma A^T)$. If we aim to include Gaussian distributions and affine maps in the same categorical structure, it is natural to study *Gaussian maps*, which are a simple kind of stochastic map consisting of an affine function with Gaussian noise, informally written $f(x) = Ax + \mathcal{N}(b, \Sigma)$, with $A \in \mathbb{R}^{n \times m}$, $b \in \mathbb{R}^n$ and $\Sigma \in \mathbb{R}^{n \times n}$. To compose this with another Gaussian map $g(y) = Cy + \mathcal{N}(d, \Xi)$, we combine their Gaussian noise independently using the transformation rule $g(f(x)) = CAx + \mathcal{N}(Cb + d, C\Sigma C^T + \Xi)$.

Definition 4.1 ((Fritz, 2020, § 6)). *The prop **Gauss** has as morphisms $m \rightarrow n$ the tuples (A, b, Σ) with $A \in \mathbb{R}^{n \times m}$, $b \in \mathbb{R}^n$ and positive semidefinite $\Sigma \in \mathbb{R}^{n \times n}$. Composition and monoidal product are given by*

$$(A, b, \Sigma); (C, d, \Xi) = (CA, Cb + d, C\Sigma C^T + \Xi)$$

$$(A_1, b_1, \Sigma_1) \oplus (A_2, b_2, \Sigma_2) = \left(\begin{pmatrix} A_1 & 0 \\ 0 & A_2 \end{pmatrix}, \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}, \begin{pmatrix} \Sigma_1 & 0 \\ 0 & \Sigma_2 \end{pmatrix} \right)$$

Gaussian distributions are the ‘states’ $0 \rightarrow n$ in **Gauss**. There is a prop morphism **AffVect** $\rightarrow$ **Gauss** representing an affine map $x \mapsto Ax + b$ as the Gaussian map $(A, b, 0)$.

4.2. Causal Graphical Quadratic Algebra and Completeness

We now turn our attention to the axiomatic theory of **Gauss**. The prop $\overline{\mathbf{GQA}}$, which stands for *causal* Graphical Quadratic Algebra, is freely generated by the following operations

$$\begin{array}{ccccccc} \text{---} \bullet & \text{---} \bullet & \text{---} \circ & \text{---} \circ & \text{---} \boxed{k} & \text{---} & \text{---} \leftarrow \end{array} \quad (8)$$

where k ranges over $\mathbb{R}$, and equations as in the first two blocks of Figure 1 (linear and quadratic fragments), *except* for axiom Z (where the generator $\text{---} \circ$ appears, which is not among those in (8)). As in related work Fritz (2020);

Cho and Jacobs (2019), the term ‘causal’ indicates that the operations in (8) should be read as processing values from left-to-right, without constraining the values of inputs; equivalently (cf. Section 3), they are *discardable*. This is the sense of ‘causal’ familiar from categorical probability, where a morphism is causal precisely when it preserves discarding Fritz (2020); Cho and Jacobs (2019)—intuitively, a causal process returns an output for every input and cannot signal information backwards. We caution that ‘causal’ carries different meanings in neighbouring fields, such as the interventionist causality of causal inference Jacobs et al. (2021), which is unrelated to the input-output reading intended here. Note we may regard $\overline{\mathbf{GQA}}$ as an extension of $\overline{\mathbf{GAA}}$ from Section 3. The extension is simply given by adding the operation $\leftarrow$ and the associated equations. It is also a fragment of full $\mathbf{GQA}$, which we will introduce later. We now define an interpretation $\langle \cdot \rangle$ of $\overline{\mathbf{GQA}}$ -generators as morphisms of $\mathbf{Gauss}$. We use the notation $f(x) = Ax + \mathcal{N}(b, \Sigma)$ for such a morphism of type $m \rightarrow n$, with $A \in \mathbb{R}^{n \times m}$ the linear component, and $b \in \mathbb{R}^n$, $\Sigma \in \mathbb{R}^{n \times n}$ defining the stochastic component (cf. Definition 4.1). Also, we write $()$ for the only element of $\mathbb{R}^0$ and $[]$ for the unique 0×0 matrix.

$$\begin{aligned} \langle \leftarrow \bullet \rangle(x) &= \begin{pmatrix} x \\ x \end{pmatrix} + \mathcal{N}\left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}\right) & \langle \leftarrow \bullet \rangle(x) &= () + \mathcal{N}(() , []) \\ \langle \bigcirc \rangle \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} &= (x_1 + x_2) + \mathcal{N}(0, 0) & \langle \bigcirc \rangle() &= (0) + \mathcal{N}(0, 0) \\ \langle \leftarrow \boxed{k} \rangle(x) &= (k \cdot x) + \mathcal{N}(0, 0) & \langle \leftarrow \rangle() &= (0) + \mathcal{N}(1, 0) & \langle \leftarrow \rangle() &= (0) + \mathcal{N}(0, 1) \end{aligned}$$

Note that the newly added operation $\leftarrow$ is the only one with a non-trivial probabilistic component.

Proposition 4.2. *The interpretation $\langle \cdot \rangle$ of $\overline{\mathbf{GQA}}$ in $\mathbf{Gauss}$ is sound, namely $c = d$ in $\overline{\mathbf{GQA}}$ implies $\langle c \rangle = \langle d \rangle$.*

PROOF. The equations in the fragment corresponding to $\overline{\mathbf{GAA}}$ are evident because $\mathbf{AffVect}$ embeds in $\mathbf{Gauss}$. It remains to show the validity of (RI) and (D). Let $R \in O(2)$, then we have

$$\langle \leftarrow \boxed{R} \rangle = \mathcal{N}(0, RR^T) = \mathcal{N}(0, I) = \langle \leftarrow \rangle \quad \langle \leftarrow \bullet \rangle = \mathcal{N}(0, 1); ! = \langle \boxed{} \rangle$$

Because of soundness, the assignment extends to a prop morphism $\langle \cdot \rangle : \overline{\mathbf{GQA}} \rightarrow \mathbf{Gauss}$, by interpreting sequential and parallel compositions of string diagrams as the corresponding operations in $\mathbf{Gauss}$ ². In fact, we can show that the equational theory of Figure 1 makes $\langle \cdot \rangle$ an isomorphism.

Theorem 4.3. *$\mathbf{Gauss}$ is presented by the fragment $\overline{\mathbf{GQA}}$.*

PROOF. For definability, let $f \in \mathbf{Gauss}(m, n)$ be given by $f(x) = Ax + \mathcal{N}(\mu, \Sigma)$. Then, using Cholesky decomposition, we can write $\Sigma = LL^T$ where L is lower triangular, and obtain as desired

$$\langle \leftarrow \begin{array}{c} \boxed{\mu} \\ \boxed{L} \\ \boxed{A} \end{array} \rangle = f. \quad (9)$$

We now tackle completeness. We may transform any string diagram of $\overline{\mathbf{GQA}}$ into the form (9), see Proposition A.10 in Appendix A.2. From this normal form, we can unambiguously read off the values of (A, μ, Σ) , following Proposition A.8. Note that a key part in the uniqueness proof is played by the property that $\leftarrow$ is invariant under arbitrary orthogonal matrices, a generalisation of RI (Proposition A.7). Finally, $\langle \cdot \rangle$ sends the normal form (9) with data (A, μ, Σ) to the Gaussian map (A, μ, Σ) ; hence equal denotations force $(A, \mu, \Sigma) = (A', \mu', \Sigma')$, yielding completeness.

5. Axiomatising Quadratic Problems

As mentioned in the introduction, Gaussian probability and quadratic problems are tightly related. We will now introduce *quadratic relations* as a categorical formalism to express quadratic problems in a compositional way. We then give a presentation $\mathbf{GQA}$ for this category, and use this presentation to study the relationship with Gaussian probability in an algebraic fashion.

²This extension is canonically defined because the prop $\overline{\mathbf{GQA}}$ is freely generated. For a comprehensive explanation of the universal property of freely generated props, which is used both here and in Section 6.1, see Baez, Coya and Rebro (2018).

5.1. Quadratic Problems as Quadratic Relations

We justify the introduction of quadratic relations in steps. The basic setup of an optimisation problem is: given a weight (or cost) function $f(x_1, \dots, x_{m+n})$, compute the parameterised infimum $(x_1, \dots, x_m) \mapsto \inf \{f(x_1, \dots, x_{m+n}) : (x_{m+1}, \dots, x_{m+n}) \in \mathbb{R}^n\}$ over some variables. We can type such a weight function as $F : \mathbb{R}^m \times \mathbb{R}^n \rightarrow [0, +\infty]$, thereby formally designating the ‘input’ and ‘output’ variables. How to compose such functions? We can see $F : \mathbb{R}^m \times \mathbb{R}^n \rightarrow [0, +\infty]$ as a *weighted relation* the same way that an ordinary relation can be seen as $R : \mathbb{R}^m \times \mathbb{R}^n \rightarrow \{0, 1\}$. This suggests a notion of composition: while relations compose as on the left below, weighted relations compose by adding the weights and minimising over the shared variable³, as on the right below:

$$(R; S)(x, z) = \bigvee_y R(x, y) \cdot S(y, z) \quad (F; G)(x, z) = \inf_y \{F(x, y) + G(y, z)\} \quad (10)$$

This way of building up optimisation problems is well-established, see e.g. Hanks, She, Patterson, Hale, Klawonn and Fairbanks (2024) for model-predictive control, Rockafellar (1997) for convex analysis (where weighted relations are called *bifunctions*), Willems (2013) for control theory, and Stein and Samuelson (2024) for a categorical perspective. In our developments, it is convenient to introduce the following notation, reminiscent of Iverson brackets (Rockafellar (1997)), associating a weighted relation $\{\!\{ \phi \}\!\}$ to a formula ϕ , denoting a linear equation between its free variables:

$$\{\!\{ \phi \}\!\} \stackrel{\text{def.}}{=} 0 \quad \text{if } \phi \text{ is true,} \quad \infty \quad \text{otherwise} \quad (11)$$

Note that 0 denotes truth and ∞ falsity in this context! The indicator function of a set A is given by $1_A(x) = \{\!\{ x \in A \}\!\}$.

Example 5.1. Consider the weighted relation $F : \mathbb{R} \times \mathbb{R}^2 \rightarrow [0, \infty]$ given by the constraint $F(c, (x, y)) = \{\!\{ c = x + y \}\!\}$, and let $G : \mathbb{R}^2 \times \mathbb{R}^0 \rightarrow [0, \infty]$ be $G((x, y), ()) = \frac{1}{2}(x^2 + y^2)$. Then the composite $F; G$ computes the parameterised quadratic problem $(F; G)(c, ()) = \inf \left\{ \frac{1}{2}(x^2 + y^2) : x + y = c \right\} = \frac{1}{4}c^2$. We will render this composition diagrammatically in **GQA**, cf. Example 5.8.

We now focus on the class of weighted relations, called *quadratic relations*, which correspond to quadratic problems: this is a type of optimisation problem involving quadratic objective functions and affine constraints. First, a *quadratic function* $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is a function that can be written as a multivariate polynomial of degree at most 2. Examples of quadratic functions are the squared euclidean distance $\|\mathbf{x}\|^2$, $(x_1 - 1)x_2$, and $x_1^2 - x_2$. A *partial quadratic function* is allowed to assume the value $+\infty$ (indicating falsity/partiality) outside an affine subspace M .

Definition 5.2 ((Rockafellar, 1997, p.109)). A partial quadratic function $f : \mathbb{R}^n \rightarrow [-\infty, \infty]$ is a function that can be written in form $f(x) = \langle x, \Sigma x \rangle + \langle b, x \rangle + c + \{\!\{ x \in M \}\!\}$ where $M \subseteq \mathbb{R}^n$ is an affine subspace, $\Sigma \in \mathbb{R}^{n \times n}$ is a symmetric matrix, $b \in \mathbb{R}^n$, $c \in \mathbb{R}$, and $\langle -, - \rangle$ denotes the standard inner product on $\mathbb{R}^n$.

We write the codomain as $[-\infty, \infty]$ to follow the standard convention for extended-real-valued (convex) functions; note, however, that a function of the above form never attains the value $-\infty$, as the term $\{\!\{ x \in M \}\!\}$ only ever adds $+\infty$. Thus partial quadratic functions take values in $\mathbb{R} \cup \{+\infty\}$.

Recall that a function $f : \mathbb{R}^n \rightarrow [-\infty, \infty]$ is *convex* if its epigraph $\{(x, y) : y \geq f(x)\}$ is a convex subset of $\mathbb{R}^{n+1}$. An *elementary convex partial quadratic function* is one in diagonal form $h(x) = \sum_i \lambda_i x_i^2$ with $\lambda_i \in [0, +\infty]$. The function h is partial, because $h(x) = \infty$ whenever $\lambda_i = \infty$ and $x_i \neq 0$.

Example 5.3. The negative log-density function of a Gaussian distribution on $\mathbb{R}^n$ is a partial convex quadratic function on $\mathbb{R}^n$. It is partial because it takes the value $+\infty$ (corresponding to density 0) outside of its support subspace S .

It turns out that nonnegative partial quadratic functions are automatically convex. Also, they are closed under the composition formula (10). We state a slightly more general proposition, namely closure under arbitrary constrained infima (see Appendix A.3 for details of the proof).

Proposition 5.4. If $M \subseteq \mathbb{R}^n \times \mathbb{R}^m$ is an affine subspace, and g is a nonnegative partial quadratic function, then so is $f(x) = \inf \{g(y) \mid (x, y) \in M\}$.

³This analogy can be made rigorous with quantales (Lawvere (1973)). Our weighted relations are parametrised over the quantale $([0, +\infty], \geq, +)$.

Definition 5.5 (Quadratic Relations). The prop **QuadRel** of quadratic relations has:

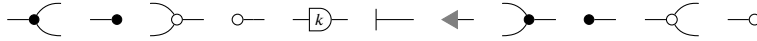
- morphisms $m \rightarrow n$ are non-negative partial quadratic functions $F : \mathbb{R}^m \times \mathbb{R}^n \rightarrow [0, \infty]$, called quadratic relations
- identities are indicator functions $\text{id}_n(x, y) = \{x = y\}$
- composition is by minimisation: for $F : m \rightarrow n$ and $G : n \rightarrow p$, define $F; G : m \rightarrow p$ as $(F; G)(x, z) := \inf_y \{F(x, y) + G(y, z)\}$.
- the monoidal product is addition. Given $F_i : m_i \rightarrow n_i$ for $i = 1, 2$, then $F_1 \oplus F_2 : m_1 + m_2 \rightarrow n_1 + n_2$ is defined by $(F_1 \oplus F_2)((x_1, x_2), (y_1, y_2)) = F_1(x_1, y_1) + F_2(x_2, y_2)$.

Composition in **QuadRel** is well-defined by Proposition 5.4, since we can write $(F; G)(x, z) = \inf\{H(\mathbf{y}) : (x, z, \mathbf{y}) \in M\}$ where $H(y_1, y_2, y_3, y_4) = F(y_1, y_2) + G(y_3, y_4)$ and $M = \{(x, z, y_1, y_2, y_3, y_4) : x = y_1, y_2 = y_3, z = y_4\}$.

Remark 5.6. To every affine relation $M \subseteq \mathbb{R}^m \times \mathbb{R}^n$ we can associate the quadratic relation 1_M given by its indicator function. This assignment respects composition, i.e. yields a prop morphism $\mathbf{AffRel} \rightarrow \mathbf{QuadRel}$; in particular, $\mathbf{AffRel}$ embeds in **QuadRel**.

5.2. Graphical Quadratic Algebra

We now introduce the diagrammatic calculus presenting **QuadRel**. The prop **GQA** is freely generated by the following operations and the equations in Figure 1, for $k \in \mathbb{R}$.



We may regard **GQA** as an extension of **GAA** from Section 3, where we add $\leftarrow$, and the associated equations. As we did for $\boxed{k}$ and $\mid$ we can define the ‘mirror image’ of $\leftarrow$ as $\rightarrow := \overleftarrow{\leftarrow}$. We now define an interpretation of **GQA**-generators as morphisms of **QuadRel**. To each operation c of type $m \rightarrow n$ we assign a quadratic relation $\llbracket c \rrbracket$ of the same type, i.e. a nonnegative partial quadratic function $\mathbb{R}^{m+n} \rightarrow [0, \infty]$. We make use of the notation $\{\emptyset\}$ introduced in (11).

$$\llbracket \text{---} \bullet \text{---} \rrbracket \left(x, \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \right) = \{x = x_1 = x_2\} \quad \llbracket \text{---} \bullet \rrbracket (x, ()) = 0 \quad \llbracket \bullet \text{---} \rrbracket ((), x) = 0$$

$$\llbracket \text{---} \cup \text{---} \rrbracket \left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, y \right) = \{y = x_1 + x_2\} \quad \llbracket \circ \text{---} \rrbracket ((), x) = \{x = 0\} \quad \llbracket \text{---} \circ \rrbracket (x, ()) = \{x = 0\}$$

$$\llbracket \text{---} \rrbracket ((), x) = \{x = 1\} \quad \llbracket \boxed{k} \text{---} \rrbracket (x, y) = \{y = k \cdot x\} \quad \llbracket \leftarrow \rrbracket ((), x) = \frac{1}{2}x^2$$

The interpretation for $\text{---} \bullet$, $\bullet \text{---}$, $\text{---} \cup$, and $\text{---} \circ$ is defined symmetrically, e.g. $\llbracket \text{---} \cup \text{---} \rrbracket \left(\begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, x \right) = \{x = x_1 = x_2\}$. Also, except for the new generator $\leftarrow$, it extends conservatively the interpretation of these operations in **AffRel** given in Bonchi et al. (2019) (see Proposition 6.2 below).

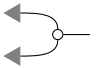
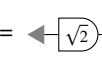
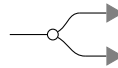
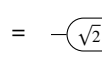
Proposition 5.7. The interpretation $\llbracket \cdot \rrbracket$ of **GQA** in **QuadRel** is sound, namely $c = d$ in **GQA** implies $\llbracket c \rrbracket = \llbracket d \rrbracket$.

PROOF. By Remark 5.6, **AffRel** embeds in **QuadRel**, so all the axioms of **GAA** are verified. It remains to verify the three axioms involving the quadratic generator, namely (RI), (D) and (Z).

$$\begin{aligned} \llbracket \leftarrow \boxed{R} \text{---} \rrbracket (\mathbf{y}) &= \inf_{\mathbf{x}} \left\{ \frac{1}{2} \|\mathbf{x}\|^2 + \|\mathbf{y} = R\mathbf{x}\| \right\} = \frac{1}{2} \|R^{-1}\mathbf{y}\|^2 = \frac{1}{2} \|\mathbf{y}\|^2 = \llbracket \leftarrow \rrbracket (\mathbf{y}) \\ \llbracket \leftarrow \bullet \rrbracket &= \inf_{\mathbf{x}} \left\{ \frac{1}{2}x^2 + 0 \right\} = 0 = \llbracket \boxed{0} \rrbracket \quad \llbracket \leftarrow \circ \rrbracket = \inf_{\mathbf{x}} \left\{ \frac{1}{2}x^2 + \|\mathbf{x} = 0\| \right\} = \frac{1}{2} \cdot 0^2 = \llbracket \boxed{0} \rrbracket \end{aligned}$$

As for $\overline{\mathbf{GQA}}$, by freeness of **GQA**, this interpretation yields a prop morphism $\llbracket \cdot \rrbracket : \mathbf{GQA} \rightarrow \mathbf{QuadRel}$.

Example 5.8. Returning to Example 5.1, $F = \llbracket \text{---} \cup \text{---} \rrbracket$, $G = \llbracket \text{---} \Rightarrow \text{---} \rrbracket$, and $\llbracket \text{---} \cup \text{---} \Rightarrow \text{---} \rrbracket = F; G$. We can solve $(F; G)(c, ()) = \inf \left\{ \frac{1}{2}(x^2 + y^2) : x + y = c \right\} = \frac{1}{4}c^2$ by equational reasoning in **GQA**. The derivation, which is

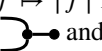
the mirror image of the one given for  =  in (3), proves  =  (observe that $\llbracket -\sqrt{2} \rrbracket(c, ()) = \frac{1}{4}c^2$).

5.3. Completeness of GQA

In this section we establish that **GQA** presents **QuadRel**. As we already proved soundness (Proposition 5.7), we focus on definability and completeness.

First, in a category where objects are equipped with Frobenius monoids, questions about arbitrary morphisms can be reduced to questions about states, i.e. morphisms $0 \rightarrow n$, as follows: for a morphism $f : m \rightarrow n$, its ‘name’ $\llbracket f \rrbracket : 0 \rightarrow m \oplus n$ is defined as

$$\llbracket f \rrbracket = \text{diagram showing a wire entering a box labeled } f \text{ from the left, and a wire exiting to the right, with a dot on the top wire} \quad (12)$$

The assignment $f \mapsto \llbracket f \rrbracket$ is bijective – every morphism can be recovered from its name by bending back the top wire, composing with  and using axiom $\bullet$ -fr and coun.

It will be useful to introduce a subprop $\overline{\mathbf{GQA}}_E$ of **GQA**, defined as follows.

Definition 5.9. The prop $\overline{\mathbf{GQA}}_E$ is freely generated as $\overline{\mathbf{GQA}}$ with the addition of the generator $\bullet$ — and equations, for $k \in \mathbb{R} \setminus \{0\}$

$$\bullet = \text{diagram of a dot}, \quad \text{diagram of a dot with a wire} \stackrel{\text{TI}}{=} \text{diagram of a dot with a wire}, \quad \text{diagram of a dot with a wire} \stackrel{\text{SI}}{=} \text{diagram of a dot with a wire}$$

Diagrams in $\overline{\mathbf{GQA}}_E$ are not only a useful intermediary step in the proof of completeness; it turns out they axiomatise the *extended Gaussians* of Stein and Samuelson (2023, 2024), which can be thought of intuitively as Gaussian maps that incorporate a form of (non-probabilistic) nondeterminism, modelled by the extra generator $\bullet$ (see Appendix A.4 for details). In this paper, we focus on **GQA**, and leave further exploration of $\overline{\mathbf{GQA}}_E$ to future work.

A fundamental piece of the completeness proof involves showing that any quadratic problem, represented as a string diagram of **GQA** with no inputs, can be turned into a string diagram of the fragment $\overline{\mathbf{GQA}}_E$. This crucial step is Theorem 5.10 below, an elimination procedure which, given a state in **GQA**, iteratively removes all occurrences of conditioning on some variable being zero ($-\circ$). The elimination procedure can therefore be understood as symbolically conditioning an (extended) Gaussian to obtain an explicit posterior, or as solving a quadratic problem via QR decomposition. We elaborate this viewpoint in Section 6.3.

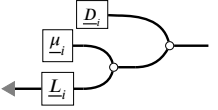
Theorem 5.10. Let $M : 0 \rightarrow n$ be a string diagram in **GQA**. Then there exists a string diagram M' in $\overline{\mathbf{GQA}}_E$, and a ‘scalar’ $\alpha : 0 \rightarrow 0$, such that $M = M' \oplus \alpha$ is derivable in **GQA**. The scalar is of the form $\llbracket \underline{c} \rrbracket$ with $c \in [0, \infty)$, or $\perp$.

Theorem 5.11. The prop **QuadRel** is presented by **GQA**.

PROOF. For definability, note that by taking names (12), it suffices to show that all states $M : 0 \rightarrow n$ (i.e. nonnegative partial quadratic functions) are definable. By Proposition A.11, we can always apply an affine change of coordinate, i.e., some invertible affine map g , such that $g(\llbracket M \rrbracket(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3)) = \frac{1}{2} \|\mathbf{x}_1\|^2 + \{\|\mathbf{x}_3\| = 0\}$, which is definable as the state

$$\text{diagram of a state } M \text{ with a dot} = \text{diagram of a state } M \text{ with a dot and a box labeled } g^{-1}.$$

For completeness, it suffices to show that for all states $M_1, M_2 : 0 \rightarrow n$ of **GQA**, if $\llbracket M_1 \rrbracket = \llbracket M_2 \rrbracket$ in **QuadRel**, then $M_1 = M_2$ is derivable in **GQA**. The steps of the proof are as follows (see Appendix A.5.2 for details). First, using Theorem 5.10, we transform each M_i (for $i = 1, 2$) into the form $\llbracket c_i \rrbracket \oplus M'_i$, i.e. $c_i \oplus M'_i$, where M'_i lies in $\overline{\mathbf{GQA}}_E$ and c_i is a leftover scalar (the *score* of M_i). Then, to each M'_i , we apply the normalisation procedure

for states in $\overline{\mathbf{GQA}}_E$ (Proposition A.13), which brings them to the form $N_i :=$ , where D_i is a

vector subspace, $\mu_i \in D_i^\perp$, L_i is lower triangular, and $\text{im}(L_i L_i^T) \subseteq D_i^\perp$. By Proposition A.22, $\llbracket N_1 \rrbracket = \llbracket N_2 \rrbracket$ if and only if $D_1 = D_2$, $\mu_1 = \mu_2$, and $L_1 L_1^T = L_2 L_2^T$. Using Proposition A.8, we can assume that $L_1 = L_2$ in the two diagrams. Then, by appealing to the completeness of **GAA**, we can show that $N_1 = N_2$. It remains to match the leftover scalars. Since the parallel composite denotes the sum $\llbracket M_i \rrbracket = \llbracket c_i \rrbracket + \llbracket N_i \rrbracket$, where $\llbracket c_i \rrbracket \in [0, \infty]$ is an additive constant, the hypothesis $\llbracket M_1 \rrbracket = \llbracket M_2 \rrbracket$ together with $\llbracket N_1 \rrbracket = \llbracket N_2 \rrbracket$ (which is finite on the support of N_i) forces $\llbracket c_1 \rrbracket = \llbracket c_2 \rrbracket$; hence $c_1 = c_2$ is derivable, as scalars with equal denotation are derivably equal (Appendix A.5.1). Therefore $M_1 = c_1 \oplus N_1 = c_2 \oplus N_2 = M_2$, as we wanted.

6. Applications

6.1. Functorial Transformations

A powerful application of our presentations for **Gauss** and **QuadRel** is the ease with which we can now define functors on these categories, as mappings on generators preserving the relevant equations. By the universal property of a presentation, such an assignment extends uniquely to a morphism of props. The first functor we consider clarifies the connection between Gaussian probability and quadratic problems. Relying on the isomorphisms $\mathbf{Gauss} \cong \overline{\mathbf{GQA}}$ and $\mathbf{QuadRel} \cong \mathbf{GQA}$, we can define a functor $L : \mathbf{Gauss} \rightarrow \mathbf{QuadRel}$ simply by mapping the generators of $\overline{\mathbf{GQA}}$ to the same generators in $\mathbf{GQA}$. Note that L is functorial by construction, even though composition operations in **Gauss** (integration/pushforward) and composition in **QuadRel** (infimisation) are at first glance very different. It remains to compute how the functor L acts concretely on a Gaussian map.

Proposition 6.1. *The functor L sends a Gaussian map $f(x) = Ax + \mathcal{N}(\mu, \Sigma)$ to the quadratic relation given by its negative conditional log-density. That is, $Lf(x, y)$ is defined as $\frac{1}{2} \langle (y - Ax - \mu), \Omega(y - Ax - \mu) \rangle + \{ (y - Ax - \mu) \in \text{im}(\Sigma) \}$, where Ω is any generalised inverse of Σ , i.e. satisfies $\Sigma \Omega \Sigma = \Sigma$.*

PROOF. The Gaussian map f is represented in $\overline{\mathbf{GQA}}$ by a string diagram of shape (9). Its interpretation in **QuadRel** is computed in detail in Proposition A.21.

Intuitively, L records a Gaussian not by its density but by the *energy* (negative log-density) of that density. Since the density of $\mathcal{N}(\mu, \Sigma)$ is proportional to $\exp(-\frac{1}{2} \langle x - \mu, \Omega(x - \mu) \rangle)$, its negative logarithm is, up to an additive constant, the quadratic form in Proposition 6.1, read as a function of the output y given the input x . The matrix Ω known as the *precision* (or *information*) matrix: when Σ is invertible it is simply $\Omega = \Sigma^{-1}$, and it measures how sharply the distribution is peaked—large precision means small variance. When Σ is singular the Gaussian is concentrated on the affine subspace $Ax + \mu + \text{im}(\Sigma)$, so Σ has no inverse; one then takes Ω to be any generalised inverse, while the term $\{ (y - Ax - \mu) \in \text{im}(\Sigma) \}$ assigns energy $+\infty$ (density 0) to points off the support. In this light, minimising Lf in y returns the most likely output, recovering the link between maximum likelihood and least squares discussed in the introduction.

Proposition 6.2. *The functor $L : \mathbf{Gauss} \rightarrow \mathbf{QuadRel}$ is faithful and $\mathbf{GQA}$ is conservative over $\overline{\mathbf{GQA}}$, meaning that $s = t$ in $\mathbf{GQA}$ implies $s = t$ in $\overline{\mathbf{GQA}}$.*

PROOF. Faithfulness is clear as every Gaussian map can be reconstructed from its log-density. For conservativity, if two $\overline{\mathbf{GQA}}$ -terms s, t are provably equal in $\mathbf{GQA}$, then $\llbracket s \rrbracket = \llbracket t \rrbracket$ in **QuadRel**. By faithfulness, $\langle s \rangle = \langle t \rangle$ in **Gauss**, hence $s = t$ is provable in $\overline{\mathbf{GQA}}$ by completeness.

These results provide an elegant and precise connection between the world of Gaussians and of convex optimisation problems. Functoriality of the log-density was laboriously proved in Stein and Samuelson (2024) by explicit means. Here, we obtain the result “for free” from our use of presentations, from which it also easily follows faithfulness and conservativity of $\mathbf{GQA}$ over $\overline{\mathbf{GQA}}$.

Another functor of interest, $S : \mathbf{Gauss} \rightarrow \mathbf{AffRel}$ can be defined by mapping every generator except $\leftarrow$ to the same generator in **AffRel**, and mapping $\leftarrow$ to $\bullet$. This functor takes a Gaussian map to the affine relation given by its support; if $f(x) = Ax + \mathcal{N}(b, \Sigma)$, then $S(f) := \{(x, y) : y \in Ax + b + \text{im}(\Sigma)\}$. Again, functoriality of this assignment is an immediate consequence of our presentation results. A similar definition yields a functor $\mathbf{QuadRel} \rightarrow \mathbf{AffRel}$, mapping $f : \mathbb{R}^{m+n} \rightarrow [0, \infty]$ to its effective domain $\text{dom } f = \{(x, y) \mid f(x, y) < \infty\}$.

6.2. Ordinary Least-Squares

We now demonstrate how to apply our theory of **GQA** to the method of ordinary least squares in linear regression. The aim of linear regression is simple: to find a linear model that best fits a set of observations. In its usual vectorial formulation, all available observations of the regressors form the columns of a single matrix A and all observations of the dependent variable form a single vector y ; then, a linear model with parameters x is expressed concisely as the system $Ax = y$. Typically, for consistency, we also assume that the regressors are linearly independent, *i.e.*, that A is injective. If A is not invertible—as it usually isn't—this system does not admit an exact solution. We can nevertheless look for parameters x such that Ax best approximates the observed values y . Here, ‘best’ is interpreted in such a way that the sum of squared errors, $\|y - Ax\|^2$, is minimised. This function can be translated directly into the following diagram:

$$\left[\begin{array}{c} \boxed{A} \boxed{-1} \\ \hline \end{array} \right] = \begin{pmatrix} x \\ y \end{pmatrix} \mapsto \frac{1}{2} \|y - Ax\|^2$$

The formula for the optimal $\hat{x}$ is well-known to be the familiar ordinary least squares (OLS) estimator $\hat{x} = (A^T A)^{-1} A^T y = A^+ y$. Our aim here is to re-derive this expression starting from the diagram above, using the axioms of **GQA**. The estimator itself is of course classical; the point of the exercise is *methodological*. It illustrates that an optimisation problem—not merely a linear-algebraic identity—can be solved entirely by equational string diagram reasoning, and that the genuinely ‘quadratic’ content is concentrated in a single step (the lemma below), the rest being pure linear algebra carried out in the fragment **GAA**. We then need a lemma which, in semantic terms, states that $\|Ax + y\|^2 = \|Ax\|^2 + \|y\|^2$ when $A^T y = 0$, *i.e.*, when y is in the kernel of the transpose of A , *i.e.*, is orthogonal to the image of A . It is easy to see why this holds: $\|Ax + y\|^2 = \|Ax\|^2 + 2\langle Ax, y \rangle + \|y\|^2$ and since $A^T y = 0$, we have $\langle Ax, y \rangle = \langle x, A^T y \rangle = 0$. We now derive it diagrammatically. Note that this is the only part of our derivation of the OLS estimator which uses the *quadratic* fragment of GQA; the rest will be plain GAA.

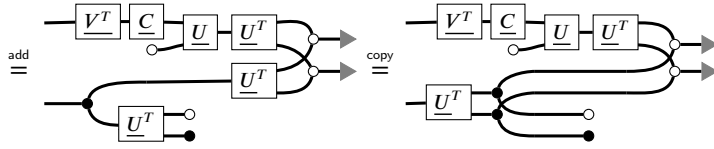
Lemma 6.3. *The following equality is derivable in GQA:*

PROOF. We will need a few useful facts from (graphical) linear algebra. First, recall that a matrix A can be written UDV^T , where U, V are orthogonal matrices and D is diagonal—this is known as the *singular value decomposition* (SVD) of A . Diagrammatically, this translates to the two equalities on the first line below (and the two on the second line for the transpose of A):

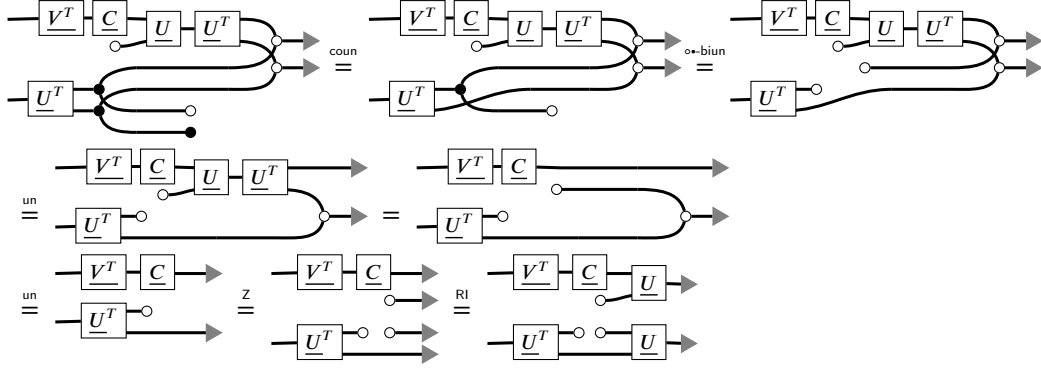
where C is the non-zero sub-matrix of D and the remaining zeros are represented by $\circ$ -. Going back to the initial least-square problem, we have

where the second equality $(*)$ holds because V is orthogonal and C is diagonal with non-zero diagonal entries (the non-zero singular values of A), so both have trivial kernel. Then, since U^T is orthogonal, we can use orthogonal invariance to obtain

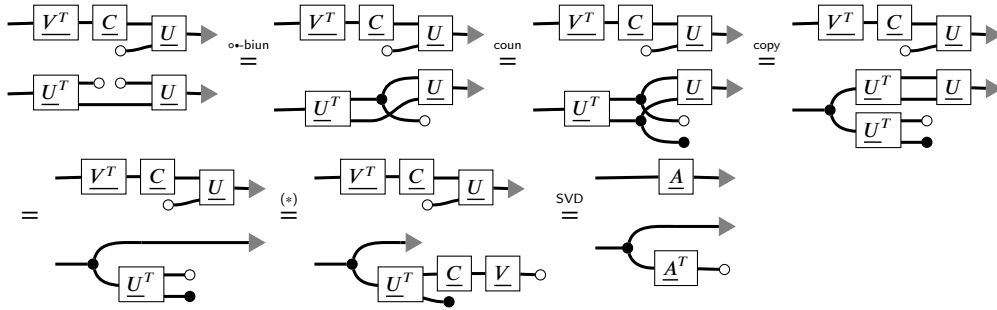
Graphical Quadratic Algebra



where the last two equalities (labelled add and copy) use the additivity of matrices (they commute with addition $\cup$) and the fact that they commute with copying $\rightarrow \bullet$ (standard facts in Graphical Affine Algebra). Then,

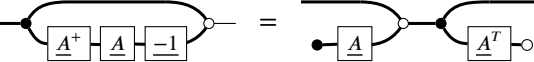


Where the last step uses again the fact that U is orthogonal to apply rotational invariance. We now repeat the same steps in the other direction:

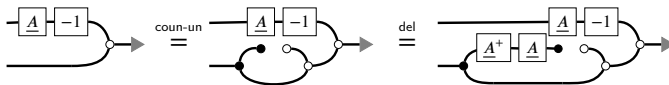


where once again, the equality $(*)$ uses that V is orthogonal and C diagonal with non-zero entries, so both have trivial kernel.

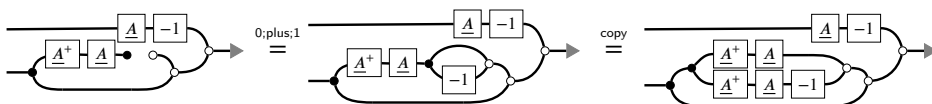
We also record the following lemma, which is just linear algebra, so we appeal to completeness of **GQA** for the proof. Intuitively, it translates the fact that $I - AA^+$ is the projection onto the orthogonal space to the range of A , i.e. onto the kernel of A^T . (Note also that AA^+ is the projection onto the range of A .)

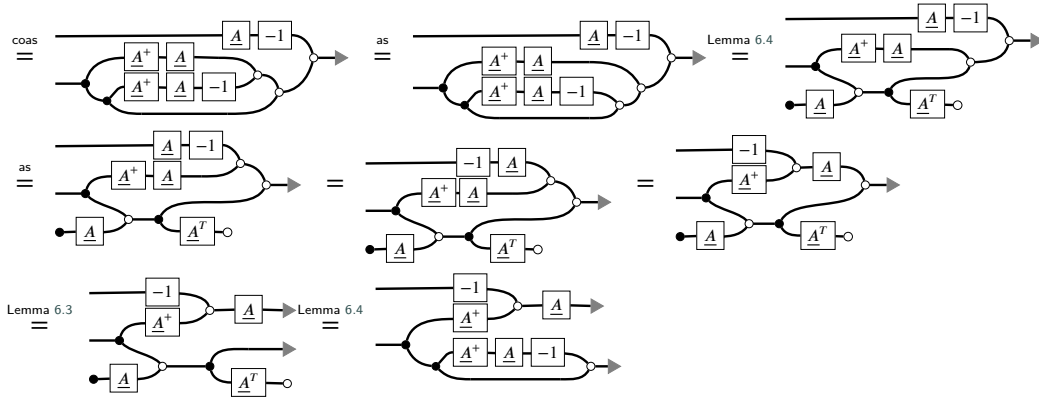
Lemma 6.4. 

We can now derive the OLS estimator. First, because $\circ -$ is the unit of $\cup$ and $\bullet -$ is the co-unit of $\rightarrow \bullet$, we have



where the second equality (del) comes from the fact that any matrix composed with $\bullet -$ is equal to $\bullet -$. Then, since $1 + (-1) = 0$, we get





The semantics of the last diagram is the quadratic function

$$\begin{pmatrix} x \\ y \end{pmatrix} \mapsto \frac{1}{2} (||AA^+y - Ax||^2 + ||y - AA^+y||^2).$$

Its infimum is clearly reached at $\hat{x} = A^+y$ as wished: in this case $||AA^+y - A\hat{x}||^2 = 0$ and the remaining term $||y - AA^+y||^2 = ||y - A\hat{x}||^2$ indicates the distance between $A\hat{x}$ and y , i.e., how far we are from having successfully inverted A .

6.3. Metatheory of Conditioning

Conditioning is the fundamental operation of statistical learning, but conditioning continuous random variables on equality is fraught with paradoxes. Even for Gaussian variables, the seemingly equivalent conditions $(X - Y) = 0$ and $(X/Y) = 1$ may produce different posteriors. This is known as Borel's paradox (e.g. Proschan and Presnell (1998)).

Our results can be understood to show that, as long as we are only working with Gaussian random variables and affine-linear transformations and conditions, Borel's paradox does *not* occur. Equivalent conditions produce equal posteriors! In fact, the meta-theory of conditioning Gaussian distributions on affine constraints is *precisely* the usual theory of those constraints, viewed as affine subspaces. For example, successively conditioning on $X = a + b$ and then $Y = X - a$ is equivalent to conditioning on $Y = b$ then $X = a + Y$, simply because the subspaces $\{(x, y) : x = a + b, y = x - a\}$ and $\{(x, y) : y = b, x = a + y\}$ are equal.

We can approach this formally as follows: because **Gauss** and **AffRel** both embed in **GQA**, we have a diagrammatic way of imposing a constraint C on a distribution ψ by composition. We can then normalize the resulting diagram using the elimination procedure and obtain back something in the fragment $\overline{\mathbf{GQA}}$ up to a constant. It remains to show why composing with the constraint in **GQA** actually computes the correct probability-theoretic posterior: interestingly, this proof can be performed purely formally, without making use of any explicit formulas for the conditionals of Gaussians; without loss of generality, we will show this for the case of conditioning a single variable on zero, i.e. composing with the constraint $\circ$.

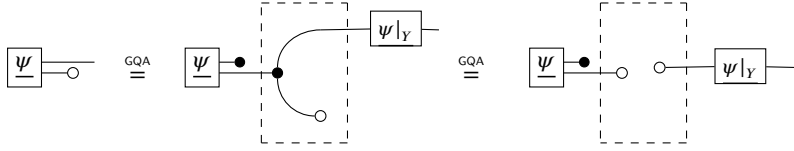
Proposition 6.5. *Let (X, Y) be multivariate Gaussian with distribution $\psi = \mathcal{N}(\mu, \Sigma)$, and let $\psi|_0$ be the conditional distribution $X|(Y = 0)$. Then $\psi|_0$ is again Gaussian, and we have a scalar term c such that*

$$\boxed{\psi} \circ \text{---} \stackrel{\text{GQA}}{=} c \boxed{\psi|_0} \text{---}$$

PROOF. Let $\psi \in \mathbf{Gauss}(0, m \oplus n)$ be the joint distribution. We use the fact that **Gauss** has conditionals in the sense of Fritz (2020). That is, there exists a morphism $\psi|_Y \in \mathbf{Gauss}(n, m)$ such that the joint distribution ψ factors as

$$\boxed{\psi} = \boxed{\psi} \bullet \begin{array}{c} \text{---} \boxed{\psi|_Y} \text{---} \\ \text{---} \end{array}$$

This is the defining property of a regular conditional probability in measure theory (c.f. (Fritz, 2020, 11.3)), i.e. $\psi|_Y$ computes the correct posterior distribution. By embedding these morphisms into **GQA**, we can now reason



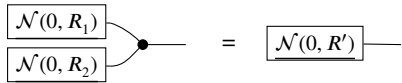
where the highlighted subdiagrams are equated by completeness for **GAA** because they both denote the subspace $\{(0, 0)\} \subseteq \mathbb{R} \times \mathbb{R}$. The composite on the right is now just $\psi|_Y$ evaluated at 0, i.e. $\psi|_0$, while the scalar on the left corresponds to the *score* of the conditioning problem, i.e. the negative logarithm of the *model evidence*.

Using the equivalence $\mathbf{GQA} \cong \mathbf{QuadRel}$, this proof also shows how the computation of Gaussian conditionals can be turned into quadratic optimisation problems, much like in probabilistic interpretations of least squares (see previous section). By embedding **Gauss** into **GQA**, we have thus internalized conditioning, which is a meta-level operation on Gaussians, into a first-class morphism in **GQA**, represented by $\multimap$. This can be used to give a convenient semantics to a small programming language centered around Gaussians and conditioning.

It is worth contrasting our $\multimap$ with conditioning as it appears in the categorical treatments of probability. In a Markov category (Fritz (2020); Cho and Jacobs (2019)), conditioning arises as *disintegration* (Bayesian inversion): a meta-level factorisation of a joint distribution into a marginal and a conditional kernel, built from causal (discardable) morphisms only. By contrast, $\multimap$ models *exact conditioning*—observing that a continuous variable takes a prescribed value, or that two variables are equal, as in the $(=:)$ operator of Stein and Staton (2021, 2024). It is a genuine effect rather than a causal morphism—indeed it lies in $\mathbf{GQA} \setminus \overline{\mathbf{GQA}}$ —and it is the operation responsible for Borel’s paradox above; the computation just given shows that, on Gaussians, it nonetheless returns the measure-theoretic regular conditional probability.

Another fundamental operation which is simple to express in **GQA** is the *multiplication* of states, that is composition with $\bowtie$. In probabilistic terms, this is *Bayesian updating*; the multiplication of distributions ψ_1 and ψ_2 combines the evidence of both distributions to obtain an even sharper prediction. This update operation lies at the core of the Kálmán filter.

Proposition 6.6. *For centered Gaussians with variance R_1, R_2 , their multiplication has variance R' with $\frac{1}{R'} = \frac{1}{R_1} + \frac{1}{R_2}$. The corresponding syntactic statement in **GQA** is*



PROOF. We can show the semantic claim by noting that the density of the multiplication is proportional to the product of densities, and thus has density proportional to

$$\exp\left(-\frac{1}{2R_1}x^2\right) \cdot \exp\left(-\frac{1}{2R_2}x^2\right) = \exp\left(-\frac{1}{2}\left[\frac{1}{R_1} + \frac{1}{R_2}\right]x^2\right) = \exp\left(-\frac{1}{2R'}x^2\right)$$

The syntactic claim follows by appealing to the completeness theorem, or directly by adapting the proof strategy of (3).

6.4. Gaussian Probabilistic Programming

In Stein and Staton (2021, 2024) the authors study a simple first-order probabilistic programming language for Gaussian probability, which we name GPL. The core probabilistic constructs are sampling from a standard-normal distribution `normal()` and an operator $(=:)$ for conditioning two random variables to be equal. Terms of GPL are, for $\alpha, \beta \in \mathbb{R}$, $i \in \{0, 1\}$,

$$s, t ::= x \mid s + t \mid \alpha \cdot s \mid \underline{\beta} \mid (s, t) \mid () \mid \text{let } x = s \text{ in } t \mid s; t \mid \pi_i s \mid \text{normal}() \mid s =: t$$

The operational semantics for GPL maintains a latent Gaussian distribution in memory over all allocated random variables. `normal()` extends that distribution with an independent normal random variable, while $(=:)$ numerically

performs conditioning and replaces the latent distribution with the posterior. The `let`-constructs binds the result of one computation to a variable, and `s; t` sequences two computations without storing the result of `s`. Pairs are constructed by (s, t) and destructed by π_1, π_2 .

We will only consider terms meaningful if they are well-typed: for example, `normal() + normal()` is well-typed but $\pi_1(\text{normal}())$ is not. The execution of well-typed programs does not get stuck, and we can assign meaningful denotational semantics to them.

Our types are generated from a single basic type R denoting *real number* or *random variable*. This is closed under pair formation and the unit type I :

$$\tau ::= R \mid I \mid \tau * \tau.$$

Thus, every type is, up to rebracketing, a power R^n for some $n \in \mathbb{N}$. A context Γ is a list of typed variables, such as $\Gamma = (x : R * R, y : R)$. In such a context, we may check well-typedness of a type of a term with free variables from Γ . For example, we have $\Gamma \vdash \pi_1(x) + y : R$. The typing judgements for GPL are standard for ML-style languages.

$$\begin{array}{c} \frac{}{\Gamma, x : \tau, \Gamma' \vdash x : \tau} \quad \frac{}{\Gamma \vdash () : I} \quad \frac{\Gamma \vdash s : \sigma \quad \Gamma \vdash t : \tau}{\Gamma \vdash (s, t) : \sigma * \tau} \quad \frac{\Gamma \vdash s : R \quad \Gamma \vdash t : R}{\Gamma \vdash s + t : R} \\[10pt] \frac{\Gamma \vdash t : R}{\Gamma \vdash \alpha \cdot t : R} \quad \frac{}{\Gamma \vdash \underline{\beta} : R} \quad \frac{}{\Gamma \vdash \text{normal}() : R} \quad \frac{\Gamma \vdash s : R \quad \Gamma \vdash t : R}{\Gamma \vdash (s := t) : I} \\[10pt] \frac{\Gamma \vdash s : I \quad \Gamma \vdash t : \tau}{\Gamma \vdash s; t : \tau} \quad \frac{\Gamma \vdash s : \sigma \quad \Gamma, x : \sigma \vdash t : \tau}{\Gamma \vdash \text{let } x = s \text{ in } t : \tau} \quad \frac{\Gamma \vdash e : \tau_1 * \tau_2}{\Gamma \vdash \pi_i e : \tau_i} \end{array}$$

Note that while addition $s + t$ of terms of type R is well-typed, multiplication is only well-typed for constants, e.g. $2 \cdot t$. This is to ensure that we are modelling Gaussian-linear models only.

We'll now give an example of an inference task expressed in GPL. Consider a noisy measurement problem: if $X \sim \mathcal{N}(0, 100)$ and $Y|X \sim \mathcal{N}(X, 25)$, what is $X|Y = 40$? In GPL, we can express this as the program

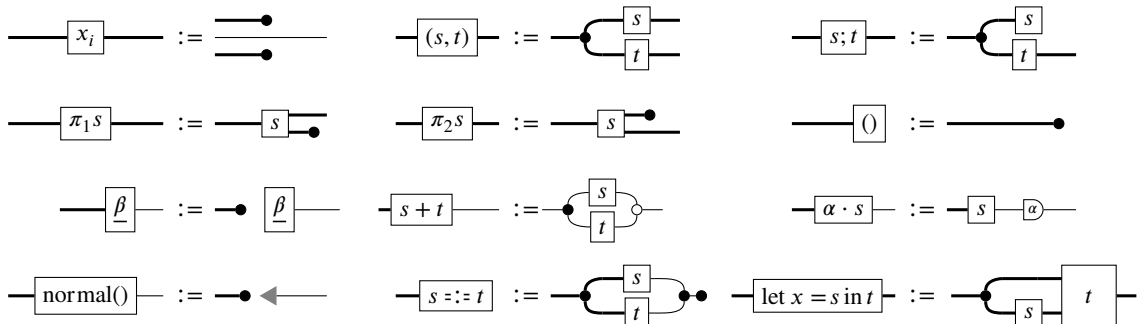
$$\vdash \text{let } x = 10 \cdot \text{normal}() \text{ in let } y = x + 5 \cdot \text{normal}() \text{ in } (y := 40); x : R \quad (13)$$

When run, this program will return the posterior distribution $\mathcal{N}(32, 20)$. We can find this answer mathematically by factorising the joint negative log-density

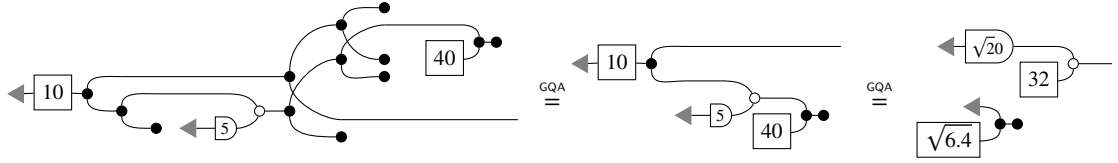
$$f(x, 40) \propto \frac{1}{2} \frac{x^2}{100} + \frac{1}{2} \frac{(x - 40)^2}{25} = \frac{1}{2} \frac{(x - 32)^2}{20} + 6.4$$

As before, the constant $c = 6.4$ corresponds to the score: the higher c , the less likely the observation was under the model. We can disregard this constant if we are only interested in the renormalized posterior.

It is possible to associate any GPL term to a **GQA**-string diagram. This translation is rigorously definable as a functor, by understanding GPL as the internal language of a suitable monoidal category, as discussed in Stein and Staton (2021); Stein (2021); Di Lavore, Jacobs and Román (2024). Given the focus of the present work, we do not pursue this systematically here, and confine ourselves to showing how the constructs of GPL map into **GQA**:



For instance, the program (13) translates to the leftmost diagram, and normalises modulo **GQA** to the posterior distribution and normalisation constant expressed by the rightmost diagram.



Via this translation, GPL receives denotational semantics in **QuadRel**, which is comparable to existing GPL semantics in interesting ways: first, the elimination procedure of Theorem 5.10 mirrors the operational semantics of Stein and Staton (2024) by reducing conditioning statements whenever they occur. Second, the normal form argument of Theorem 5.11 can be also employed to compare two GPL terms once understood as **GQA** diagrams. This may yield a decidability procedure for contextual equivalence, and thus a fully abstract (and completely axiomatised) denotational semantics for GPL, which unlike the one in Stein and Staton (2021) does not rely on an equivalence relation that is hard to decide. A complete account of this conjecture is outside of the scope of this paper; we leave it for future work.

6.5. Resistor Networks and Connections to Symplectic Algebra

We now connect our development to a body of recent work on compositional accounts of electrical and quantum systems (Baez and Fong (2015); Booth, Carette and Comfort (2024b,a)). To illustrate this, we will show how resistor networks (a simple class of electrical circuits) can be embedded in **GQA**. We will also consider a different embedding of a larger class of circuits into **GQA** in Section 6.6. The treatment in this section follows Baez and Fong (2015).

A resistor network is a simple type of electrical circuit composed only of fixed Ohmic resistors. Such a network can be understood as a labelled graph $G = (V, E)$ where some nodes have been marked as external boundary nodes ∂V . It turns out that the external behavior of such a circuit is completely captured by its *power functional* $Q : \mathbb{R}^{\partial V} \rightarrow \mathbb{R}$ which describes how much power is dissipated when potentials $\phi \in \mathbb{R}^{\partial V}$ are placed at the boundary. For example, the power functional associated with a single resistor of resistance R is

$$Q(\phi_0, \phi_1) = \frac{1}{2R}(\phi_0 - \phi_1)^2 \quad (14)$$

Here the boundary consists of the resistor's two terminals, $\partial V = \{0, 1\}$, so a boundary potential is a pair $\phi = (\phi_0, \phi_1) \in \mathbb{R}^{\partial V} \cong \mathbb{R}^2$ and we write $Q(\phi_0, \phi_1)$ for $Q(\phi)$ with its single argument spelled out in components. When circuits are composed, the potentials at their internal nodes are chosen in a way to minimize the overall power functional: this is the “principle of least power”, which is equivalent to the usual Kirchhoff laws governing voltages and currents in the circuit.

Example 6.7 (Resistors in sequence Baez and Fong (2015)). When plugging two resistors R_1 and R_2 in sequence, the power functional of the composite circuit is given by the quadratic optimization problem

$$Q(\phi_0, \phi_2) = \inf_{\phi_1} \{Q_1(\phi_0, \phi_1) + Q_2(\phi_1, \phi_2)\} = \inf_{\phi_1} \left\{ \frac{1}{2R_1}(\phi_0 - \phi_1)^2 + \frac{1}{2R_2}(\phi_1 - \phi_2)^2 \right\}$$

which is solved by the power functional of the resistor $R = R_1 + R_2$:

$$Q(\phi_0, \phi_2) = \frac{1}{2(R_1 + R_2)}(\phi_0 - \phi_2)^2$$

It is shown in Baez and Fong (2015) that the power functionals of electrical circuits are precisely the *Dirichlet forms*, i.e. quadratic functions that can be written in the form

$$Q(\phi) = \sum_{ij} \frac{1}{2} c_{ij} (\phi_i - \phi_j)^2 \text{ for } c_{ij} \geq 0$$

Dirichlet forms are also closed under composition (minimisation), however they do not form a category because there is no Dirichlet form that behaves as an identity morphism (which would correspond to an ideal wire of zero resistance).

However, by embedding Dirichlet forms into the larger category of quadratic relations we can obtain a compositional account of resistor networks. In [Baez and Fong \(2015\)](#), the authors use a different embedding to obtain a category, namely Lagrangian subspaces: we return to this at the end of the section.

Having seen that we can describe resistor networks as quadratic relations via their energy functionals, we will now study this interpretation in a syntactic way using **GQA**. Using the Gaussian interpretation of **GQA**, we will then obtain a nice dictionary to translate between resistor networks and Gaussian probabilistic models.

The semantics of the resistor (14) is easy to describe.

$$Q(\phi_0, \phi_1) = \left[\begin{array}{c} \boxed{\mathcal{N}(0, R)} \\ \text{---} \end{array} \right] (\phi_0, \phi_1) \quad (15)$$

We introduce the shorthand $\text{---} \overset{R}{\circ} \text{---}$ for this term. The following equations are then provable in **GQA**.

Example 6.8 (Resistors in sequence). *We have*

$$\text{---} \overset{R_1}{\circ} \text{---} \overset{R_2}{\circ} \text{---} = \left[\begin{array}{c} \boxed{\mathcal{N}(0, R_1)} \\ \text{---} \end{array} \right] \left[\begin{array}{c} \boxed{\mathcal{N}(0, R_2)} \\ \text{---} \end{array} \right] \text{---} = \left[\begin{array}{c} \boxed{\mathcal{N}(0, R_2)} \\ \boxed{\mathcal{N}(0, R_1)} \end{array} \right] \text{---} = \text{---} \overset{R_1 + R_2}{\circ} \text{---}$$

This equation has the following simple reading in **Gauss**: if $\phi_1 \sim \mathcal{N}(\phi_0, R_1)$ and $\phi_2 \sim \mathcal{N}(\phi_1, R_2)$ then $\phi_2 \sim \mathcal{N}(\phi_0, R_1 + R_2)$.

Example 6.9 (Resistors in parallel). *We have*

$$\begin{array}{c} \text{---} \overset{R_1}{\circ} \text{---} \\ \text{---} \overset{R_2}{\circ} \text{---} \end{array} = \left[\begin{array}{c} \boxed{\mathcal{N}(R_1)} \\ \text{---} \end{array} \right] \left[\begin{array}{c} \boxed{\mathcal{N}(R_2)} \\ \text{---} \end{array} \right] \text{---} = \left[\begin{array}{c} \boxed{\mathcal{N}(R_1)} \\ \boxed{\mathcal{N}(R_2)} \end{array} \right] \text{---} = \text{---} \overset{R'}{\circ} \text{---}$$

where $\frac{1}{R'} = \frac{1}{R_1} + \frac{1}{R_2}$. The dotted equation in **GAA** can be established using the completeness theorem by checking that both sides denote the same subspace

$$\{(x_1, y_1, x_2, y_2) : x_1 + y_1 = x_2 + y_1 = y_2\} = \{(x_1, y_1, x_2, y_2) : x_1 = x_2 \wedge x_1 + y_1 = y_2\}$$

It remains to apply the multiplication formula for Gaussians ([Proposition 6.6](#)).

These examples suggest that resistors and their composition organise into a monoidal structure which can be encoded in **GQA**, with functorial semantics in **QuadRel**. To make this formal, let us introduce a prop **Res** freely generated by the signature

$$\text{---} \overset{R}{\circ} \text{---} \quad \text{---} \left[\begin{array}{c} \text{---} \\ \text{---} \end{array} \right] \text{---} \quad \text{---} \bullet \quad \left[\begin{array}{c} \text{---} \\ \text{---} \end{array} \right] \bullet \quad \text{---} \bullet$$

where the parameter R ranges over the positive reals, and equations:

$$\begin{array}{l} \text{---} \left[\begin{array}{c} \text{---} \\ \text{---} \end{array} \right] \text{---} = \text{---} \left[\begin{array}{c} \text{---} \\ \text{---} \end{array} \right] \text{---} \quad \text{---} \bullet = \text{---} \quad \left[\begin{array}{c} \text{---} \\ \text{---} \end{array} \right] \bullet = \left[\begin{array}{c} \text{---} \\ \text{---} \end{array} \right] \bullet \quad \left[\begin{array}{c} \text{---} \\ \text{---} \end{array} \right] \left[\begin{array}{c} \text{---} \\ \text{---} \end{array} \right] \text{---} = \left[\begin{array}{c} \text{---} \\ \text{---} \end{array} \right] \text{---} \quad \left[\begin{array}{c} \text{---} \\ \text{---} \end{array} \right] \bullet = \text{---} \\ \left[\begin{array}{c} \text{---} \\ \text{---} \end{array} \right] \left[\begin{array}{c} \text{---} \\ \text{---} \end{array} \right] \text{---} = \left[\begin{array}{c} \text{---} \\ \text{---} \end{array} \right] \text{---} \quad \left[\begin{array}{c} \text{---} \\ \text{---} \end{array} \right] \left[\begin{array}{c} \text{---} \\ \text{---} \end{array} \right] \bullet = \left[\begin{array}{c} \text{---} \\ \text{---} \end{array} \right] \bullet \quad \left[\begin{array}{c} \text{---} \\ \text{---} \end{array} \right] \left[\begin{array}{c} \text{---} \\ \text{---} \end{array} \right] \bullet = \text{---} \quad \left[\begin{array}{c} \text{---} \\ \text{---} \end{array} \right] \left[\begin{array}{c} \text{---} \\ \text{---} \end{array} \right] \left[\begin{array}{c} \text{---} \\ \text{---} \end{array} \right] \text{---} = \text{---} \overset{R}{\circ} \text{---} \end{array} \quad (16)$$

The equations, apart from the last, are those of special Frobenius monoids. The final equation reflects the fact that resistors are bidirectional. Morphisms $m \rightarrow n$ of Res represent resistor networks with m open terminals on the left, assembled from individual resistors by connecting them in series (sequential composition) and in parallel (possibly sharing terminals through the Frobenius structure).

The power-functional semantics of (14) then organises into a morphism of props $\mathcal{R} : \text{Res} \rightarrow \mathbf{GQA}$, which maps the generators $\text{---}\overset{R}{\circ}\text{---}$ as in (15), sequential and parallel composition according to Examples 6.8 and 6.9, and the Frobenius structure of Res to that of $\mathbf{GQA}$. Consequently $\mathcal{R}$ exhibits resistor networks as a sub-prop of $\mathbf{GQA}$, and the power functional of any resistor network Res is recovered as the partial quadratic function denoted by its image $\llbracket \mathcal{R}(r_R) \rrbracket$.

We may now return to the comparison with related work. As remarked earlier, Baez and Fong (2015) wish to form a category of quadratic functions akin to **QuadRel**, but notice the lack of identity morphisms. While our work uses *partial* quadratic functions to remedy this, their work uses a specific encoding of quadratic functions as Lagrangian subspaces. In particular, if $Q : \mathbb{R}^V \rightarrow \mathbb{R}$ is a quadratic function then its gradient $\nabla Q : \mathbb{R}^V \rightarrow \mathbb{R}^V$ is linear, so we obtain a linear subspace

$$\Gamma(Q) = \{(\psi, \nabla Q(\psi)) : \psi \in \mathbb{R}^V\} \subseteq \mathbb{R}^{2V} \quad (17)$$

Such linear subspaces can be characterised, in terms of a symplectic structure on $\mathbb{R}^{2V}$, as being *Lagrangian*, giving rise to the notion of a *Lagrangian relation*, much like the linear and affine ones. Graphical calculi for Lagrangian relations have been developed by Booth et al. (2024b,a), in a diagrammatic style closely related to ours. In particular Booth et al. (2024a) explicitly features a notion of *Gaussian relation* that closely matches ours. As their approach differs in applications (quantum computation), presentation, and the use of complex numbers, we shall leave further comparison between the partial quadratic function approach taken here and the Lagrangian relation approach to future work.

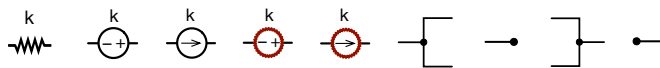
6.6. Stochastic Systems à la Willems

The mathematical analysis of stochastic processes also plays an important role in control theory. Whereas traditional systems theory studies the stochastic evolution of closed systems, Willems' behavioural approach argues that a relational formalism, akin to quadratic relations, is better suited to model the interaction of a system with its environment Willems (2013); Polderman and Willems (1998). One of Willems' motivating examples is *noisy electrical circuits*. Thermal noise, also called Johnson-Nyquist noise, expresses the fact that a resistor may produce current even when no voltage is applied. Its stochastic representation is as a Gaussian ϵ_V with zero mean and standard deviation σ . For example, Willems discusses the Ohmic resistor with thermal noise displayed on the left in (18) below, and argues that it has an equivalent representation, as an Ohmic resistor in parallel with a random current source, on the right in (18) below.



The equivalence of the two circuits is established at a semantic level: the leftmost expresses the relation $V = rI + \epsilon_V$ between voltage V , resistance r , current I , and noise ϵ_V , whereas the rightmost expresses $I = V/r + \epsilon_I$, with ϵ_I the gaussian with zero mean and standard deviation σ/r .

Thanks to **GQA**, we can now give a purely equational proof of this equivalence. First, similarly to the previous section, we establish a functorial interpretation of noisy electrical circuits in **GQA**. The prop ECirc_N of noisy electrical circuits is defined as the one freely generated by a signature and equations as follows. The signature features generators



where the parameter k ranges over the non-negative reals, and equations as in (16) (with $\text{---}\overset{k}{\text{---}}$ replacing $\text{---}\overset{R}{\circ}\text{---}$). Note voltage and current sources are duplicated because the second set, with indented red boundaries, represent their 'noisy' counterpart (in a way made precise by the semantics given below). Morphisms $m \rightarrow n$ of ECirc_N represent open noisy linear electrical circuits with m open terminals on the left and n open terminals on the right.

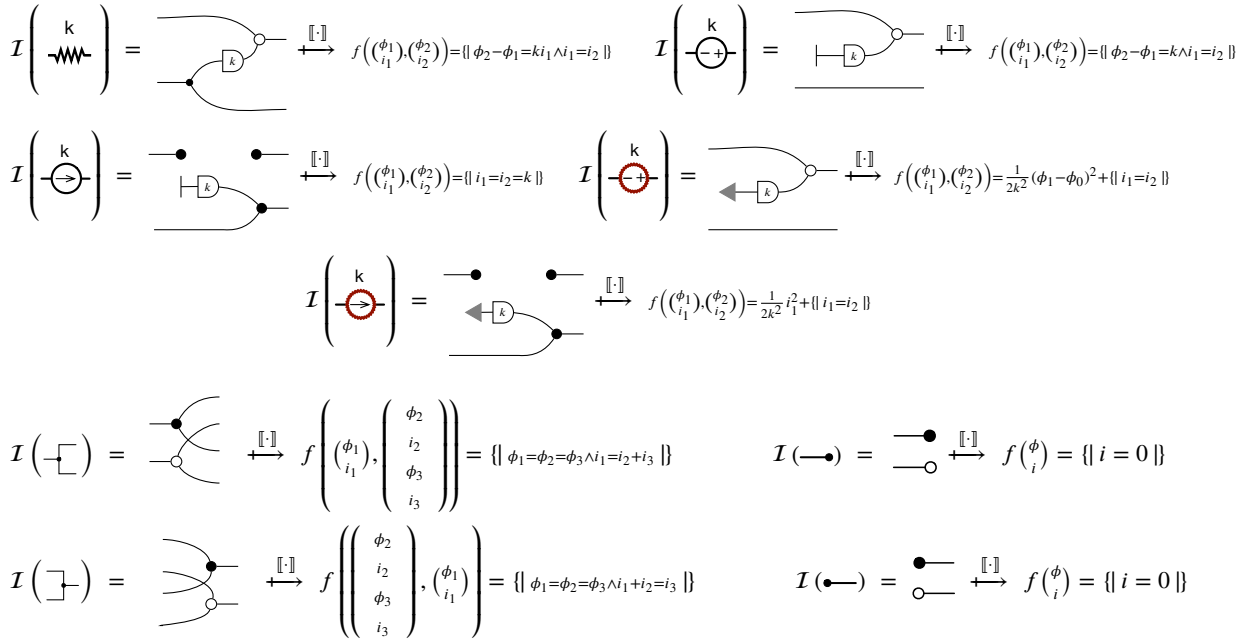


Figure 3: Functorial semantics of noisy electrical circuits. $\phi \in \mathbb{R}$ ranges over potentials and $i \in \mathbb{R}$ ranges over currents.

We can then define a functorial interpretation $\mathcal{I}(\cdot) : \text{ECirc}_N \rightarrow \mathbf{GQA}$ as in Figure 3. It is a conservative extension of the interpretation of (non-noisy) electrical circuits as string diagrams for Graphical Affine Algebra (Bonchi et al. (2019)), where we use the extra generator $\leftarrow$ to model thermal noise. Analogously to that work, a circuit of type $m \rightarrow n$ is mapped to a string diagram of $\mathbf{GQA}$ of type $2m \rightarrow 2n$: each terminal of the circuit is represented by *two* wires of the diagram, one carrying the electric potential (voltage) ϕ at that terminal and one carrying the current i flowing through it. This way, a component may receive semantics via $\text{ECirc}_N \rightarrow \mathbf{GQA} \rightarrow \mathbf{QuadRel}$ as the quadratic relation between such voltage–current pairs, with the behaviour of the individual elements given by Kirchhoff’s laws (Figure 3). We discuss this more extensively, also in comparison with Section 6.5, in Remark 6.10 below. This least-squares view is easily reconciled with Willems’ formulation in terms of open stochastic systems: using the normalisation procedure of Theorem 5.10, any string diagram with interpretation in $\mathbf{QuadRel}$ is equivalent (modulo rewiring) to one with interpretation in $\mathbf{GaussEx}$ (cf. Appendix A.4), allowing us to switch from the least-squares perspective to the stochastic-system perspective.

Remark 6.10 (The two encodings of a resistor). *The (noiseless) resistor generator of ECirc_N is the same device studied in Section 6.5, seen through the gradient duality discussed there. In Section 6.5 a resistor is encoded by its power functional $Q(\phi_0, \phi_1) = \frac{1}{2R}(\phi_0 - \phi_1)^2$ on a single wire per terminal, carrying only the potential; here it denotes the relation $\phi_2 - \phi_1 = k i$ on two wires, carrying voltage and current (the figure’s parameter k being the resistance R). The two are exchanged by differentiation: the current is the gradient of the power, $i = \partial Q / \partial \phi_0 = \frac{1}{R}(\phi_0 - \phi_1)$, so the voltage–current relation used here is exactly the graph $\Gamma(Q)$ of the gradient of the power functional, the Lagrangian subspace of (17). The single-wire energy encoding and the two-wire behavioural encoding are thus two faces of the same resistor, related by the Legendre/symplectic duality of Section 6.5.*

We are now ready to return to Willems' example of the noisy resistor, as in (18). Via the interpretation of Figure 3, the two circuits of Willems' example can be represented as string diagrams in **GQA** as follows.

(19)

We can now prove their equivalence with an equational derivation in **GQA**, as follows:

7. Conclusions

We gave a compositional account of quadratic optimisation via a category of quadratic relations called **QuadRel**, and provided a complete calculus **GQA** to reason algebraically about this class of problems. We also showed that a fragment **GQA** of **GQA** axiomatises Gaussian probabilistic maps, thus highlighting the relationship between Gaussian probability and convex analysis. We illustrated our approach by discussing functorial translations between the Gaussian, quadratic, and affine domain, by modelling the method of ordinary least-squares in linear regression as diagrammatic reasoning, by giving semantics to probabilistic programming, and by modelling various families of circuits — a development that connects our work with Willems' approach to open stochastic system in control theory (Willems (2013)). These applications are by no means intended to be the final word on the subject, but rather a starting point for a more systematic treatment. In particular, we plan to give a complete account of full abstraction for Gaussian probabilistic programming, as outlined at the end of Section 6.

From an algebraic viewpoint, we would like to explore whether the symmetries present in Graphical Affine Algebra extend to Graphical Quadratic Algebra. In particular, we conjecture the 'colour-swap' symmetry, mapping eg. $\text{---}\text{---}\text{---}$ to $\text{---}\text{---}\text{---}$ and $\text{---}\text{---}\text{---}$ to $\text{---}\text{---}\text{---}$, extends to **GQA** by mapping $\text{---}\text{---}\text{---}$ to itself. The semantics of this functor should correspond to convex conjugation (Legendre transformation). Indeed, in convex analysis terminology, every quadratic relation is a so-called convex bifunction (Rockafellar (1997)), which have been studied in categorical terms in Stein and Samuelson (2024). Independently and at the same time as a preprint (Stein, Zanasi, Piedeleu and Samuelson (2024)) of our conference paper Stein et al. (2025), the authors of Booth et al. (2024a) introduced a notion of 'Gaussian relation' in the context of quantum computation; their semantics is based on Lagrangian relations instead of partial quadratic functions, but their

axiomatisation is highly reminiscent of **GQA**. Furthermore, their presentation exhibits the colour-swap symmetry discussed earlier, which is interpreted by the Fourier transform. The precise relationship between these approaches remains to be explored. Another interesting point to explore concerns the expressivity of the calculus. Our results have been stated over the field $\mathbb{R}$, and our compact axiomatisation using rotation invariance crucially relies on the existence of square roots of nonnegative elements. On the other hand, Baez and Fong (2015) have formulated their theory of Dirichlet forms over arbitrary ordered fields, and graphical affine algebra is known to be complete over arbitrary fields and even semirings (Bonchi et al. (2019)). On the side of extending the field, the work of Booth et al. (2024a) discusses complex-valued generalizations of Gaussians as does Comfort and Lemay (2025). It would be interesting future work to delineate over which ordered fields or rings our presentation remains complete.

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PROOF. Without loss of generality we can assume that $m_1 = m_2$: this is because if, say, $m_1 > m_2$, we can extend B with further zero columns to $\tilde{B} = (B \ 0)$ and this still satisfies the assumptions $\bullet \text{---} \tilde{B} \text{---} = \bullet \text{---} B \text{---} \circ \text{---} b_0 = \bullet \text{---} B \text{---} = \bullet \text{---} A \text{---}$ of the proposition. By column equivalence (Proposition A.1), $AR = B$ for some invertible matrix R . Then the desired equality follows from the fact that $\bullet \text{---}$ is invariant under invertible matrices (Proposition A.3).

Proposition A.5. *If S, D are complementary subspaces, then we can prove in **GAA** that*

$$\begin{array}{c} \boxed{D} \\ \text{---} \end{array} \text{---} = \begin{array}{c} \boxed{D} \\ \text{---} \\ \boxed{P_S} \end{array} \text{---}$$

where P_S, P_D are the projection operators onto the subspaces.

PROOF. Using the fact that $P_S + P_D = I$, we can derive in **GAA**.

$$\begin{aligned} \bullet \text{---} \boxed{P_D} \text{---} &= \bullet \text{---} \begin{array}{c} \boxed{P_D} \\ \boxed{P_D} \\ \boxed{P_S} \end{array} \text{---} \\ &= \bullet \text{---} \begin{array}{c} \boxed{P_D} \\ \boxed{P_D} \\ \boxed{P_S} \end{array} \text{---} \quad (*) \quad \bullet \text{---} \begin{array}{c} \boxed{P_D} \\ \boxed{P_S} \end{array} \text{---} \\ \text{TI} \quad &= \bullet \text{---} \begin{array}{c} \boxed{P_D} \\ \boxed{P_S} \end{array} \text{---} = \bullet \text{---} \begin{array}{c} \boxed{P_D} \\ \boxed{P_S} \end{array} \text{---} \end{aligned}$$

In the step (*) we use the fact that matrices are linear (commute with $\text{---} \text{---}$), which is provable inside **GAA**.

A.2. Completeness of $\overline{\text{GQA}}$

Proposition A.6. $\text{---} \text{---}$ is flip invariant, i.e.

$$\text{---} \boxed{-1} \text{---} = \text{---} \text{---}$$

PROOF. By discardability and rotation invariance for $\varphi = \pi$,

$$\text{---} = \text{---} \bullet = \text{---} \begin{array}{c} \boxed{-1} \\ \boxed{-1} \end{array} \bullet = \text{---} \boxed{-1} \text{---}$$

We show here that the fragment $\overline{\text{GQA}}$, as introduced in Section 4.2, presents **Gauss**. The key is a normal form argument, which builds on the following propositions. The first proposition generalises axiom RI to arbitrary orthogonal matrices and the second will allow us to derive the uniqueness of our normal form.

Proposition A.7. *For any $n \times n$ orthogonal matrix R we can derive in $\overline{\text{GQA}}$ that $\text{---} \boxed{R} \text{---} = \text{---}$.*

PROOF. A *Givens rotation* is a rotation matrix which acts as a rotation along two coordinate axes, and the identity otherwise. From (RI), it follows immediately that --- is invariant under Givens rotations. Now we use the fact (proved below) that every orthogonal matrix R can be written as a product $R = Q_1 \cdots Q_n \cdot D$ where the Q_i are Givens rotations and D is a diagonal matrix with entries ± 1 . From this, the proposition follows by repeated application of invariance under Givens rotations and flip invariance (Proposition A.6).

It remains to prove the claimed decomposition of R , which we establish as follows: the usual algorithm for QR-decomposition gives $R = QU$, where Q is a product of Givens rotations and U is upper triangular. Then $U = Q^T R$ is orthogonal (normal) in addition to being upper triangular, hence must be diagonal, with entries ± 1 . All of these linear algebraic transformations can be performed diagrammatically in $\overline{\text{GQA}}$, by Proposition 3.2.

Proposition A.8. For $A \in \mathbb{R}^{n \times m_1}$, $B \in \mathbb{R}^{n \times m_2}$ such that $AA^T = BB^T$, $\leftarrow \boxed{A} \text{---} = \leftarrow \boxed{B} \text{---}$.

PROOF. Without loss of generality we can assume that $m_1 = m_2 = m$: this is because if, say, $m_1 > m_2$, we can extend B with further zero columns to $\tilde{B} = (B \ 0)$ and this still satisfies the assumptions $AA^T = \tilde{B}\tilde{B}^T$ and

$$\leftarrow \boxed{\tilde{B}} \text{---} = \leftarrow \boxed{B} \text{---} \stackrel{(D)}{=} \leftarrow \boxed{B} \text{---} = \leftarrow \boxed{A} \text{---}$$

By Proposition A.2, $A = BR$ for some $R \in O(m)$, hence $\leftarrow \boxed{A} \text{---} = \leftarrow \boxed{B} \text{---}$ follows from Proposition A.7.

We can now derive an encoding of Gaussian distributions in $\overline{\mathbf{GQA}}$.

Definition A.9. Let $\mathcal{N}(\mu, \Sigma)$ be a Gaussian distribution on $\mathbb{R}^n$. Then we define

$$\boxed{\mathcal{N}(\mu, \Sigma)} \text{---} := \leftarrow \begin{array}{c} \boxed{\mu} \\ \boxed{L} \end{array} \text{---}$$

whenever we have a factorization $\Sigma = LL^T$. Such a factorization always exists by Cholesky decomposition of Σ , and by Proposition A.8 the resulting diagram is independent of the choice of L , modulo derivable equality in $\mathbf{GQA}$.

Proposition A.10. Let $M : m \rightarrow n$ be any string diagram in $\overline{\mathbf{GQA}}$. Then M can be brought into the form

$$\begin{array}{c} \boxed{\mathcal{N}(\mu, \Sigma)} \\ \boxed{A} \end{array} \text{---}$$

with $A \in \mathbb{R}^{n \times m}$, $\mu \in \mathbb{R}^n$ and $\Sigma \in \mathbb{R}^{n \times n}$ positive semidefinite.

PROOF. By the laws of symmetric monoidal categories, (Figure 2) it is possible to move all the occurrences of the generators $\leftarrow : 0 \rightarrow 1$ to the extreme left of any given string diagram where they occur, while preserving equality in $\overline{\mathbf{GQA}}$. Applying this to M , we obtain a string diagram of shape $\leftarrow \boxed{\tilde{M}} \text{---}$ equivalent to M , where $\tilde{M}$ lies in the affine fragment $\mathbf{AffVect}$ of $\overline{\mathbf{GQA}}$, meaning that $\langle \tilde{M} \rangle$ is given by $f(x_1, x_2) = Lx_1 + \mu + Ax_2$ for some matrices $A, L \in \mathbb{R}^{n \times m}$ and $\mu \in \mathbb{R}^n$. Now, $\langle \tilde{M} \rangle$ is also a morphism of $\mathbf{AffVect}$: by definition of $\langle \tilde{M} \rangle$ as above and completeness of $\overline{\mathbf{GAA}}$ for the affine fragment Bonchi et al. (2019), we have that

$$\leftarrow \boxed{\tilde{M}} \text{---} = \leftarrow \begin{array}{c} \boxed{L} \\ \boxed{\mu} \\ \boxed{A} \end{array} \text{---}$$

is provable in $\overline{\mathbf{GAA}}$, and thus in $\overline{\mathbf{GQA}}$. We can then conclude by observing that, by definition of composition in $\mathbf{Gauss}$ and functoriality of $\langle \cdot \rangle$:

$$\leftarrow \begin{array}{c} \boxed{L} \\ \boxed{\mu} \\ \boxed{A} \end{array} \text{---} = \begin{array}{c} \boxed{\mathcal{N}(\mu, LL^T)} \\ \boxed{A} \end{array} \text{---}$$

A.3. Quadratic Relations

This section is devoted to showing Proposition 5.4. We will use the following characterisation.

Proposition A.11. The following are equivalent for a function $f : \mathbb{R}^n \rightarrow [-\infty, \infty]$

1. f is a nonnegative partial quadratic function

2. f can be written in the form $f(x) = h(A(x - a)) + c$ where A is an invertible matrix, $c \geq 0$ and h an elementary convex partial quadratic function
3. f can be written as $f(x) = \inf \{ ||y||^2 : (x, y) \in M \}$ where $M \subseteq \mathbb{R}^n \times \mathbb{R}^m$ is an affine subspace.

Let us first describe the meaning of these statements. Item 2 is an analogue of Sylvester's law of inertia for partial quadratic functions; in fact, it suffices to allow the values $\lambda_i \in \{0, 1, \infty\}$ on the diagonal. Item 3 shows that nonnegative partial convex functions are precisely the class of functions that arise in constrained least squares optimisation [Boyd and Vandenberghe \(2004\)](#), which justifies our choice of using them as the morphisms of the semantic domain **QuadRel**.

PROOF (PROOF OF PROPOSITION A.11). (1) $\Rightarrow$ (2) using an affine coordinate change, we can assume that the domain of f is the subspace $M = \mathbb{R}^m \times \{0\}$. The restricted function $\tilde{f}(x_1, \dots, x_m) = f(x_1, \dots, x_m, 0, \dots, 0)$ is a total nonnegative quadratic function so we can apply the characterisation [Proposition A.11](#) to $\tilde{f}$ to obtain the desired form

$$f(x) = h(A((x_1, \dots, x_m) - a)) + \infty \cdot x_{m+1} + \dots + \infty \cdot x_n + c$$

(2) $\Rightarrow$ (3) Under the affine coordinate change $\mathbf{y} = A(\mathbf{x} - \mathbf{a})$, it suffices to consider $f(\mathbf{y}) = h(\mathbf{y}) + c$ where h is an elementary quadratic function of the form $h(\mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3) = ||\mathbf{y}_1||^2 + \{ \mathbf{y}_3 = 0 \}$. In this case, we can define the affine relation

$$M = \{(\mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3), (\mathbf{y}_1, \sqrt{c}) \mid \mathbf{y}_3 = 0\}$$

and have $f(x) = \inf \{ ||z||^2 : (x, z) \in M \}$.

(3) $\Rightarrow$ (1) If $M \subseteq \mathbb{R}^n \times \mathbb{R}^m$ is an affine relation, then there exists a vector subspace $D \subseteq \mathbb{R}^n$ and an affine map $g : \mathbb{R}^n \rightarrow D^\perp$ such that

$$M_x = \begin{cases} g(x) + D, & x \in \pi_X M \\ \emptyset, & \text{otherwise} \end{cases}$$

where $M_X = \{y : (x, y) \in M\}$. Thus we have

$$\inf \{ ||y||^2 : y \in M_x \} = ||g(x)||^2 + \{ x \in \pi_X M \}$$

which is evidently a nonnegative partial quadratic function.

We are now ready for the proof of [Proposition 5.4](#).

PROOF (PROOF OF PROPOSITION 5.4). Let $g : \mathbb{R}^m \rightarrow [-\infty, \infty]$ be a nonnegative partial quadratic function and $M \subseteq \mathbb{R}^m \times \mathbb{R}^n$ an affine relation. Using the representation of [Proposition A.11](#) and an affine coordinate change, it suffices to consider the case where g is an elementary function, i.e. a constrained minimisation problem of the form

$$f(x) = \inf \{ ||\mathbf{y}_1||^2 + \{ \mathbf{y}_3 = 0 \} : (x, (\mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3)) \in M \}$$

We define the affine relation $M' = \{(x, \mathbf{y}_1) : (x, (\mathbf{y}_1, \mathbf{y}_2, 0)) \in M\}$ and conclude that $f(x) = \inf \{ ||y||^2 : (x, y) \in M' \}$ which is a partial quadratic function by [Proposition A.11](#).

A.4. Completeness for Extended Gaussian Processes

An *extended Gaussian distribution* is a Gaussian distribution on a quotient space $\mathbb{R}^n/D$; the vector subspace D is called a *nondeterministic fibre*. This concept intends to model the idea of a distribution carrying both a probabilistic and a nondeterministic component. We can write it in additive notation $\mathcal{N}(\mu, \Sigma) + D$, where we think of the subspace D as an idealised uniform distribution. Extended Gaussians support the same basic transformations as ordinary Gaussians, such as pushforwards and conditioning [Stein and Samuelson \(2023\)](#). A related notion appears in [Willems' approach to control theory](#), under the name of open stochastic system [Willems \(2013\)](#). A major reason of interest for extended Gaussians is that they give semantics to Gaussian probabilistic programming extended with a construct `uniform()` to sample from a uniform prior [Stein and Samuelson \(2023, 2024\)](#).

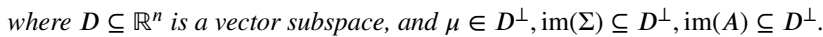
Similarly to Gaussian maps ([Section 4](#)), one may define a notion of extended Gaussian map, informally written as $f(x) = Ax + \mathcal{N}(\mu, \Sigma) + D$, where $Ax + \mathcal{N}(\mu, \Sigma)$ is a Gaussian map and D is the nondeterministic component. A **prop GaussEx** of these maps is defined [Stein and Samuelson \(2023\)](#).

We first recall the complete definition of the **prop GaussEx** of extended Gaussian maps.

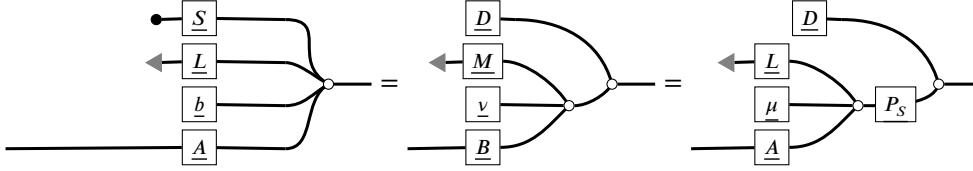
The monoidal product of $(m \xrightarrow{A} k \xleftarrow{P} n, \varphi)$ and $(m' \xrightarrow{A'} k' \xleftarrow{P'} n', \varphi')$ is defined as $(m + m' \xrightarrow{A \oplus A'} k + k' \xleftarrow{P \oplus P'} n + n', \varphi \oplus \varphi')$, using the monoidal product in **Vect** on the cospan legs and the one in **Gauss** on the distributions. The sequential composition of $(m_1 \xrightarrow{A_1} k_1 \xleftarrow{P_1} n, \varphi_1)$ and $(n \xrightarrow{A_2} k_2 \xleftarrow{P_2} n_2, \varphi_2)$ is defined as $(m_1 \xrightarrow{A_1; L} k \xleftarrow{P_2; R} n_2, \varphi)$, where $k_1 \xrightarrow{L} k \xleftarrow{R} k_2$ is the pushout of $k_1 \xleftarrow{P_1} n \xrightarrow{A_1} k_2$ in **Vect**, in **Vect** and φ is the convolution in **Gauss** of distributions $\varphi_1; L: 0 \rightarrow k$ and $\varphi_2; R: 0 \rightarrow k$.⁴

$$f(x, y) = \langle Py - Ax - b, \Sigma^+(Py - Ax - b) \rangle + \{ \| Py - Ax - b \in \text{im}(\Sigma) \| \}.$$

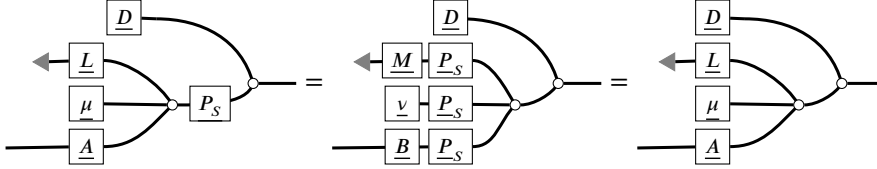
Proposition A.13. *Let $M : m \rightarrow n$ be any string diagram in $\overrightarrow{\mathbf{GQA}}_E$. Then M can be brought into the form*

Stein, Zanasi, Piedeleu, Samuelson: *Preprint submitted to Elsevier*

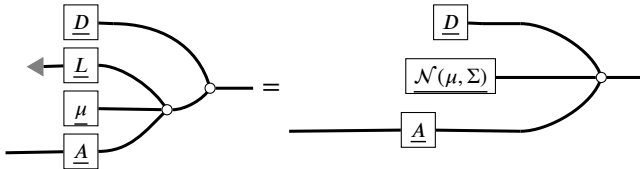
Then, let $D := \text{im}(S)$. By the associativity of $\curvearrowright$ — followed by Proposition A.5, we get



Then, by additivity of matrices (they commute with the sum $\curvearrowright$), we have



where P_S is the orthogonal projection onto $D^\perp$, and $A := P_S B$, $L := P_S M$, $\mu = P_S v$. Finally, with $\Sigma := LL^T$, we get



Using similar methods as in Section 4.2, we can now show that the prop $\overline{\mathbf{GQA}}_E$ presents **GaussEx**.

Theorem A.14. *The prop **GaussEx** is presented by $\overline{\mathbf{GQA}}_E$.*

An intuitive reading of Theorem A.14 is that the extra generator $\bullet$ — uniformly ‘outputs’ values in $\mathbb{R}$, and thus suffices to model the form of nondeterminism expressed by the component D in the definition of extended Gaussians.

First, we need to define an interpretation $\langle\langle \cdot \rangle\rangle$ of the generators of $\overline{\mathbf{GQA}}_E$ in **GaussEx**:

$$\begin{aligned} \langle\langle \bullet \rangle\rangle &:= \left(0 \xrightarrow{Id} 0 \xleftarrow{!} 1, \mathcal{N}(\cdot, [\cdot]) \right) \\ \langle\langle \boxed{g} : m \rightarrow n \rangle\rangle &:= \left(m \xrightarrow{A} n \xleftarrow{Id} n, \mathcal{N}(b, \Sigma) \right) \end{aligned}$$

for any $\overline{\mathbf{GQA}}$ -generator $\boxed{g}$ with $\langle \boxed{g} \rangle = x \mapsto Ax + \mathcal{N}(b, \Sigma)$, where $! : 1 \rightarrow 0$ is the only linear map $x \mapsto ()$. This interpretation extends to a prop morphism $\overline{\mathbf{GQA}}_E \rightarrow \mathbf{GaussEx}$, which we also write $\langle\langle \cdot \rangle\rangle$.

We now turn attention to the proof of Theorem A.14.

PROOF (PROOF OF THEOREM A.14). The proof consists in verifying definability, soundness, and completeness under the prop morphism $\langle\langle \cdot \rangle\rangle$. First we show definability. Let $f = (m \xrightarrow{A} k \xleftarrow{P} n, \mathcal{N}(\mu, \Sigma))$ be a morphism of **GaussEx**, informally representing the map $f(x) = Ax + \mathcal{N}(\mu, \Sigma) + D$ where $D = \ker(P)$. Then f is definable by the term

$$\begin{array}{c} \boxed{D} \\ \curvearrowright \\ \boxed{\mathcal{N}(\mu, \Sigma)} \\ \curvearrowright \\ \boxed{A} \end{array} \quad (20)$$

For completeness, we note that the form (20) is not quite a normal form for **GaussEx** semantics, but it becomes one if we insist on the further conditions that

$$\mu \in D^\perp, \text{im}(A) \subseteq D^\perp, \text{im}(\Sigma) \subseteq D^\perp$$

where $D^\perp \subseteq \mathbb{R}^n$ is the orthogonal complement of D . As detailed in Proposition A.13, we can transform every string diagram of $\overline{\mathbf{GQA}}_E$ into that specific shape. The values of D , A , μ , Σ are then uniquely determined from the interpretation $\langle\langle \cdot \rangle\rangle$ in **GaussEx**.

A.5. Completeness of GQA

A.5.1. Scalars

In this section we study properties of scalars ($0 \rightarrow 0$ morphisms of **GQA**). We use the following shorthand notation:

$$\begin{array}{c} \text{) } \\ \bullet \end{array} := \begin{array}{c} \text{) } \\ \bullet \end{array} \quad \begin{array}{c} \text{) } \\ \circ \end{array} := \begin{array}{c} \text{) } \\ \circ \end{array} \text{ } \circ$$

Proposition A.15. *For $a, b, c \in [0, \infty)$, we have*

$$\begin{array}{c} \boxed{c} \\ \text{) } \\ \bullet \end{array} = \begin{array}{c} \boxed{c} \\ \text{) } \\ \circ \end{array} \quad \text{and} \quad \begin{array}{c} \boxed{a} \\ \text{) } \\ \bullet \end{array} \begin{array}{c} \boxed{b} \\ \text{) } \\ \bullet \end{array} = \begin{array}{c} \boxed{c} \\ \text{) } \\ \bullet \end{array}$$

whenever $a^2 + b^2 = c^2$.

PROOF. The first equation follows from flip invariance of $\leftarrow$ (Proposition A.6). For the second equation, we can find a rotation matrix $R \in \mathbb{R}^{2 \times 2}$ which sends the vector (a, b) to $(c, 0)$. Then by (RI) and (Z)

$$\begin{array}{c} \boxed{a} \\ \text{) } \\ \bullet \end{array} \begin{array}{c} \boxed{b} \\ \text{) } \\ \bullet \end{array} = \begin{array}{c} \boxed{\begin{pmatrix} a \\ b \end{pmatrix}} \\ \text{) } \\ \bullet \end{array} = \begin{array}{c} \boxed{\begin{pmatrix} a \\ b \end{pmatrix}} \\ \text{) } \\ \bullet \end{array} \begin{array}{c} \boxed{R^T} \\ \text{) } \\ \bullet \end{array} \\ = \begin{array}{c} \boxed{\begin{pmatrix} a \\ b \end{pmatrix}} \\ \text{) } \\ \bullet \end{array} \begin{array}{c} \boxed{R} \\ \text{) } \\ \bullet \end{array} = \begin{array}{c} \boxed{\begin{pmatrix} c \\ 0 \end{pmatrix}} \\ \text{) } \\ \bullet \end{array} \\ = \begin{array}{c} \boxed{c} \\ \text{) } \\ \bullet \end{array} \begin{array}{c} \text{) } \\ \circ \end{array} = \begin{array}{c} \boxed{c} \\ \text{) } \\ \bullet \end{array} \begin{array}{c} \text{) } \\ \circ \end{array} \begin{array}{c} \text{) } \\ \circ \end{array} = \begin{array}{c} \boxed{c} \\ \text{) } \\ \bullet \end{array}$$

Proposition A.16. *The equation*

$$\begin{array}{c} \boxed{a} \\ \text{) } \\ \bullet \end{array} = \begin{array}{c} \boxed{b} \\ \text{) } \\ \bullet \end{array}$$

is derivable in **GQA** if and only if $a^2 = b^2$.

PROOF. If $a^2 = b^2$, then $a = \pm b$, and the equation can be derived from flip invariance of $\leftarrow$ (Proposition A.6). In **QuadRel**, their denotations are given by $\frac{1}{2}a^2, \frac{1}{2}b^2$ respectively.

The initialisation principle states that if $X \sim \mathcal{N}(0, 1)$ and we condition X to be equal to a constant c , the posterior distribution is precisely c up to a scalar.

Proposition A.17 (Initialisation Principle). *The following schema is derivable for all $c \in [0, \infty)$.*

$$\begin{array}{c} \boxed{c} \\ \text{) } \\ \bullet \end{array} \text{ --- } = \begin{array}{c} \boxed{c} \\ \text{) } \\ \bullet \end{array} \begin{array}{c} \bullet \\ \text{) } \\ \bullet \end{array} \begin{array}{c} \boxed{c} \\ \text{) } \\ \bullet \end{array} \text{ --- } \quad (\text{INI})$$

PROOF. This is given by the derivation

$$\begin{array}{c} \boxed{c} \\ \text{) } \\ \bullet \end{array} \text{ --- } = \begin{array}{c} \boxed{c} \\ \text{) } \\ \bullet \end{array} \begin{array}{c} \text{) } \\ \bullet \end{array} \begin{array}{c} \text{) } \\ \bullet \end{array} \begin{array}{c} \boxed{c} \\ \text{) } \\ \bullet \end{array} \text{ --- } = \begin{array}{c} \boxed{c} \\ \text{) } \\ \bullet \end{array} \begin{array}{c} \boxed{c} \\ \text{) } \\ \bullet \end{array}$$

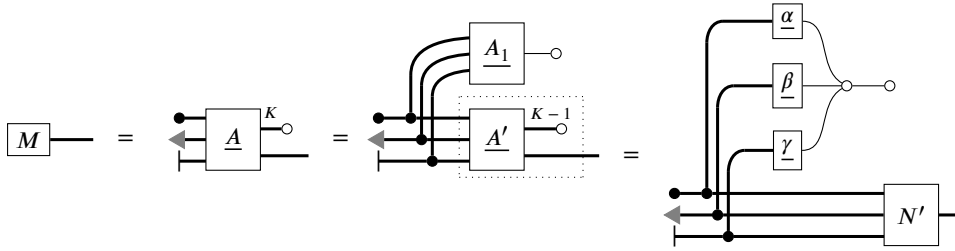
where in the first step we use the black Frobenius algebra structure, and in the second step we use axioms (1-dup) and (dup) to cancel $\text{---} \begin{array}{c} \text{) } \\ \bullet \end{array}$.

A.5.2. Elimination Procedure for Conditioning

Let $M : 0 \rightarrow n$ be a state in **GQA**. We give an inductive procedure to remove occurrences of the conditioning effect $\multimap$, except possibly in the scalar ∞ .

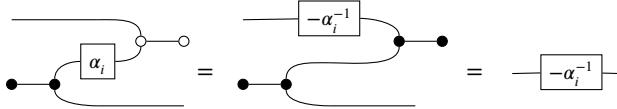
Theorem 5.10. *Let $M : 0 \rightarrow n$ be a string diagram in **GQA**. Then there exists a string diagram M' in $\overline{\mathbf{GQA}}_E$, and a 'scalar' $\alpha : 0 \rightarrow 0$, such that $M = M' \oplus \alpha$ is derivable in **GQA**. The scalar is of the form $\boxed{c} \multimap$ with $c \in [0, \infty)$, or $\multimap$.*

PROOF. We proceed by induction over the number K of generators $\multimap$ appearing in M . First, by the laws of symmetric strict monoidal categories, (Figure 2) it is possible to move all the occurrences of the generators of type $0 \rightarrow 1$ ($\multimap$, $\bullet$, and $\vdash$) to the extreme left of any given string diagram of **GQA** where they occur, while preserving equality of string diagram in the equational theory. Similarly, we can move all occurrences of $\multimap$ to the right. The result of applying this to M is displayed in the first equality below. The remaining string diagram $\underline{A}$ is necessarily in the affine fragment of **GQA**, its semantics being some matrix A . By applying equational reasoning in **GQA** (in fact, in its fragment $\overline{\mathbf{GAA}}$), which is complete for **AffVect** Bonchi et al. (2019), we can then separate A into blocks, obtaining a diagram for its first row A_1 , and one for the remaining rows A' as on the right-hand side of the second equality below. Finally, we can break down A_1 further into row vectors α , β , and γ , for each of the columns corresponding to the three incoming wires, as on the right-hand side of the third equality below:

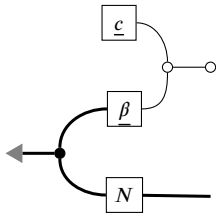


where N' is the diagram in the dotted box above, which contains $K - 1$ occurrences of $\multimap$.

We now distinguish two cases. If in α there is a coefficient $\alpha_i \neq 0$, we eliminate $\multimap$ using equational reasoning in $\overline{\mathbf{GAA}}$, as follows:



Note that this step at the same time eliminates one occurrence of the generator $\bullet$. In the remaining case, all α_i s are equal to 0, then by equational reasoning in **GQA** we can discard α and the wire going into it. We can simplify the string diagram further, first by repeatedly applying axiom (1-dup) to $\vdash$, then by including the remaining occurrences of $\bullet$ in N' , which yields N . This brings to a string diagram in the shape

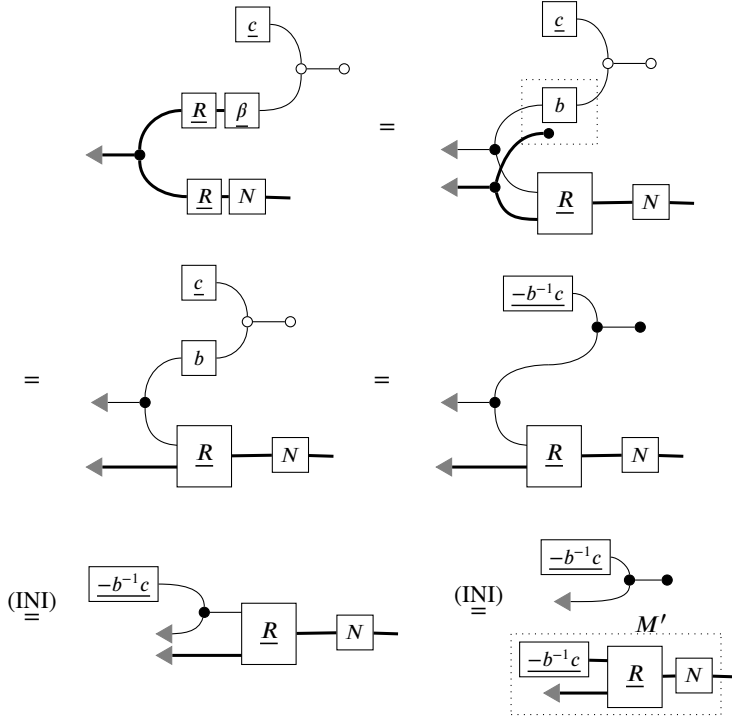


with $c \in \mathbb{R}$. We can now find an orthogonal matrix $R \in O(n)$ with $\beta R = (b, \dots, 0)$ where $b = \|\beta\|$:

$$\begin{array}{c} \boxed{c} \\ \vdash \\ \boxed{\beta} \\ \vdash \\ \bullet \\ \vdash \\ \boxed{N} \end{array} \stackrel{\text{RI}}{=} \begin{array}{c} \boxed{c} \\ \vdash \\ \boxed{\beta} \\ \vdash \\ \bullet \\ \vdash \\ \boxed{N} \end{array} \stackrel{\text{RI}}{=} \begin{array}{c} \boxed{c} \\ \vdash \\ \boxed{R} \\ \vdash \\ \bullet \\ \vdash \\ \boxed{R} \\ \vdash \\ \boxed{N} \end{array} \quad (21)$$

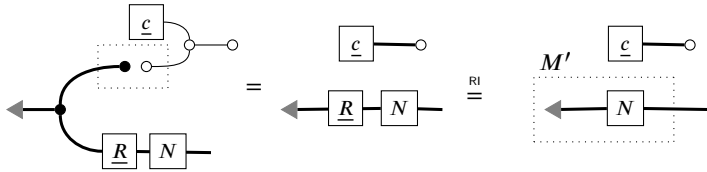
where we have used the fact that matrices commute with copying $\multimap$ (i.e., can be copied). Now, since $\beta R = (b, 0, \dots, 0)$ there are two further cases to consider.

- If $b > 0$, we can simplify the rightmost diagram of (21) further, with the matrix $\beta R = (b, 0, \dots, 0)$ reducing to the string diagram in the dotted box in the first step below (cf. (7)). Then, using (INI) as defined in Proposition A.17, we get:



where the last dotted frame identifies the diagram M' that we wanted to prove the statement of the theorem.

- If $b = 0$, the rightmost string diagram of (21) instead simplifies as



where the scalar is either 0 or ∞ depending on whether $c = 0$.

Then the scalars are normalised separately according to the procedure of Appendix A.5.1.

We now give further lemmas needed for the proof of completeness of **GQA**.

First, recall that a *generalised inverse* of a matrix $A \in \mathbb{R}^{n \times m}$ is a matrix $\Omega \in \mathbb{R}^{m \times n}$ such that $A\Omega A = A$; that is, for all $y \in \text{im}(A)$ we have that Ωy is a preimage of y . Every matrix has a generalised inverse, but it is generally not uniquely defined. The *Moore-Penrose pseudoinverse* A^+ is a particularly well-known and well-behaved choice of generalized inverse. It appears in the explicit solution to the least-squares problem:

$$\inf \{ \|x\|^2 : Ax = y \}$$

which is attained at $x = A^+ y$ when $y \in \text{im}(A)$.

Proposition A.18. *Let $f : m \rightarrow n$ be a quadratic relation and $c \in \mathbb{R}^n$, then*

$$\left[\begin{array}{c} \underline{c} \\ \underline{f} \end{array} \right] (x, y) = f(x, y - c)$$

PROOF. We compute

$$\inf_{y_1, y_2} \{ f(x, y_1) + \{ y_2 = c \} + \{ y = y_1 + y_2 \} \} = f(x, y - c)$$

Proposition A.19. In **QuadRel**, we have the following interpretations for each constant $c \in \mathbb{R}^n$, matrix $A \in \mathbb{R}^{n \times m}$ and affine subspace $W \subseteq \mathbb{R}^n$

$$\llbracket c \rrbracket (x) = \{ x = c \} \quad \llbracket A \rrbracket (x, y) = \{ y = Ax \} \quad \llbracket W \rrbracket (x) = \{ x \in W \}.$$

Proposition A.20. Let $A \in \mathbb{R}^{n \times m}$, then

$$\llbracket \leftarrow \boxed{A} \rrbracket (y) = \frac{1}{2} \|A^+ y\|^2 + \{ y \in \text{im}(A) \}$$

where A^+ is the Moore-Penrose pseudoinverse of A . This expression is furthermore equal to

$$\frac{1}{2} \langle y, \Omega y \rangle + \{ y \in \text{im}(A) \}$$

where Ω is any generalized inverse of AA^T .

PROOF. The denotation corresponds to the quadratic problem

$$\inf_x \left\{ \frac{1}{2} \|x\|^2 + \{ y = Ax \} \right\} = \inf \left\{ \frac{1}{2} \|x\|^2 : y = Ax \right\}$$

which is solved using the Moore-Penrose pseudoinverse. We have

$$\langle A^+ y, A^+ y \rangle = \langle y, (A^+)^T A^+ y \rangle = \langle y, \Omega y \rangle$$

As a Corollary of Proposition A.20 and Proposition A.18, we can compute the following semantics.

Proposition A.21. In **QuadRel**, we have

$$\llbracket \mathcal{N}(\mu, \Sigma) \rrbracket (x) = \frac{1}{2} \langle (x - \mu), \Omega(x - \mu) \rangle + \{ (x - \mu) \in \text{im}(\Sigma) \}$$

where Ω is any generalized inverse of Σ .

We can now show the following normal form result:

Proposition A.22. Consider subspaces $D_1, D_2 \subseteq \mathbb{R}^n$, and data $\mu_1 \in D_1^\perp, \mu_2 \in D_2^\perp, \text{im}(\Sigma_1) \subseteq D_1^\perp, \text{im}(\Sigma_2) \subseteq D_2^\perp$ such that the following two denotations in **QuadRel** agree:

$$\llbracket \begin{array}{c} \boxed{D_1} \\ \mathcal{N}(\mu_1, \Sigma_1) \end{array} \rrbracket = \llbracket \begin{array}{c} \boxed{D_2} \\ \mathcal{N}(\mu_2, \Sigma_2) \end{array} \rrbracket \quad (22)$$

Then $D_1 = D_2, \mu_1 = \mu_2$ and $\Sigma_1 = \Sigma_2$.

PROOF. Let $f_1 : \mathbb{R}^n \rightarrow [0, \infty]$ denote the quadratic relation denoted by the right hand side of (22), and $g_1 = \llbracket \mathcal{N}(\mu_1, \Sigma_1) \rrbracket$ as computed in Proposition A.21. We know that by assumption

$$g(x_1) < \infty \Rightarrow x_1 \in \mu_1 + \text{im}(\Sigma_1) \Rightarrow x_1 \in D_1^\perp$$

Thus in the following infimisation we must choose $x_1 \in D_1^\perp$, and thus the projections $x_1 = P_{D^\perp}(x), x_2 = P_D(x)$.

$$\begin{aligned} f_1(x) &= \inf_{x_1, x_2} \{ g(x_1) + \{ x_2 \in D_1 \} + \{ x_1 + x_2 = x \} \} \\ &= \inf_{x_1, x_2} \{ g(x_1) + \{ x_1 \in D_1^\perp \} + \{ x_2 \in D_1 \} + \{ x_1 + x_2 = x \} \} \\ &= g_1(P_{D^\perp}(x)) \end{aligned}$$

The quadratic function f_1 is thus degenerate (i.e. periodic) in the direction of D_1 . We claim that furthermore the quadratic function g_1 is nonperiodic along $D_1^\perp$, i.e. for $x_2 \in D_1^\perp$, if $\forall x(g_1(x + x_2) = g_1(x))$ then $x_2 = 0$:

- if $x_2 \in D_1^\perp \setminus \text{im}(\Sigma_1)$ then $g_1(\mu_1) = 0$ but $g_1(\mu_1 + x_2) = \infty$ so x_2 is not a period
- if $x_2 \in \text{im}(\Sigma_1)$ then x_2 can only be a period if $x_2 \in \ker(\Omega)$, but $\ker(\Omega) = \ker(\Sigma^T) = \text{im}(\Sigma)^\perp$ thus $x_2 = 0$.

We can therefore recover D_1 precisely as the kernel of f_1

$$D_1 = \{x_2 : \forall x(f_1(x) = f_1(x + x_2))\}$$

Restricting f_1 along $D_1^\perp$ furthermore recovers g_1 , i.e. the denotation of the Gaussian distribution in Proposition A.21, from which μ_1 and Σ_1 can be determined. If we define f_2, g_2 analogously for the right-hand side of (22) then from the assumption $f_1 = f_2$, we obtain $D_1 = D_2$ and $g_1 = g_2$, thus $\mu_1 = \mu_2, \Sigma_1 = \Sigma_2$ as desired.